

Observation of continuous time crystals and quasi-crystals in spin gases

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In this work, Huang and colleagues report the observation of continuous time crystal (CTC) and quasi-crystal (CTQC) phases in a noble-gas nuclear spin system (Xe-129) coupled to Rb-87 spins. By introducing a gradient field and feedback coupling between the Rb and Xe nuclear spins, the system exhibits several dynamical phases: limit cycle, quasi-periodicity, and chaos, with additional signatures of spontaneous time-translation symmetry breaking. The manuscript also discusses the “melting” of these time-ordered phases into chaotic motion and their eventual re-formation at stronger gradient fields.

In my opinion, the topic is timely, echoing with the growing interest in nonequilibrium phases of matter and time-crystal behavior. The experimental approaches using spin-exchange collisions, feedback loops, and gradient fields provide a novel platform to investigate such non-equilibrium many-body phenomena and could potentially motivate further exploration. However, I am not fully convinced by several key aspects in the current manuscript. So I would suggest the authors to address the following comments before I can make a recommendation for the publication at Nature Communications.

Major Points

1. The derivation of equation (1) and underlying Hamiltonian: Equation (1) (the set of nonlinear Bloch equations) is the central part of the paper's argument. However the authors provide limited detail on how it is derived from a the Hamiltonian governing the system interaction. It is not immediately clear how interactions, pumping, feedback, and approximations enter into these effective equations. I would recommend the author to provide a detailed derivation by starting from the relevant spin Hamiltonian in a magnetic field. This derivation can go into the Methods/Supplementary Materials. However in the main text, it would be good to provide a high-level intuition on how to obtain the Eqn.1 from the full quantum Hamiltonian.
2. Related to 1, my current understanding is that the nonlinear Bloch equations presented resemble a classical spin system rather than a quantum system, i.e. $P(w)$ in eqn.1 is a number, rather than operator. Is this understanding correct? If so, could the authors elaborate on this a bit more. Seems that recent studies have shown that time crystal behavior should also persist in classical spin systems.
3. Here the main novelty is that the authors observe quasi-periodic oscillations with multiple incommensurate frequencies, i.e. continuous time quasicrystal. However, it is not clear at the moment what key ingredients in their experimental system allow the observation of this quasi-crystal behavior. In particular, compared to the cited works on CTC using other experimental platforms, could the author discuss more explicitly what are the unique ingredients (e.g., the role of the gradient field, feedback strength, etc.) that enable this quasi-crystal behavior?
4. Time-Window Analysis: The authors mention that time-ordered phases (CTC or CTQC) only persist over a certain “stationary window,” which becomes narrower if one measures longer (since this is a continuous situation, one does not need to worry about the lifetime of the spins). The “melting” into chaos is super interesting, but it is not clear how to distinguish in a robust way between the different dynamical regimes (e.g. for Fig. 2h and 2i, without the lines, it is not immediately obvious to me the difference. Could the authors provide additional experimental with a few varying time windows to show how the system transitions from time-ordered oscillations to chaotic dynamics.
5. The feedback strength α and the field gradient g are core tunable parameters in the experiments. While Fig 3 and 4

shows the effect of tuning both α and g , it would be helpful if the authors could provide a more systematic exploration of the (α , g) parameter space to show a phase diagram-like plot of the CTC CTQC behaviors.

Minor Comment

6. The manuscript briefly describes how the system transitions from a time-crystalline (CTC or CTQC) phase into a chaotic one and back again ("re-formation"). These abrupt transitions appear to be quite interesting to me. However, there are not too much discussions around it. Could the authors comment more on this, e.g. provide some intuition on such behavior, and when this will appear and how does this depend on α and g .

Reviewer #2

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Reviewer #3

(Remarks to the Author)

Review of "Observation of Continuous Time Crystals and Quasi-Crystals in Spin Gases"

The authors report on the experimental observation of Continuous Time Crystals (CTCs) and Continuous Time Quasi-Crystals (CTQCs) in noble-gas nuclear spins. Utilizing a system of ^{129}Xe spins coupled to optically pumped ^{87}Rb atoms, they introduce an external feedback mechanism that sustains persistent oscillations of nuclear spin polarization. The feedback induces nonlinear interactions between the spins, leading to rich dynamical phases, including:

- Limit cycle oscillations, identified as CTCs, characterized by persistent, periodic behavior with random time phases.
- Quasi-periodic oscillations, associated with CTQCs, where the system exhibits multi-frequency behavior akin to spatial quasi-crystals.
- Chaotic oscillations, interpreted as the "melting" of time crystals.

The paper highlights the robustness of these phases against noise perturbations and the observation of an unusual reverse phase transition from chaos back to limit cycles. The authors claim that these findings open new avenues in precision measurements, such as multimode masers and quantum metrology.

The manuscript is well-written and offers a fresh perspective by bridging concepts from nonlinear dynamics and condensed matter physics. However, several concerns arise regarding the interpretation of the results and the technical depth of the analysis.

1. The authors attribute the random phase of oscillations in repeated runs to spontaneous breaking of continuous time-translation symmetry, framing it as evidence of time crystal behavior. However, this random phase behavior is a well-known property of self-oscillating systems with feedback. It is a natural consequence of the system's initial conditions and nonlinear dynamics, rather than a unique signature of a new phase of matter.

2. The robustness of the oscillations against noise perturbations is attributed to the intrinsic properties of time crystals. However, this robustness is more accurately explained by the nonlinear negative feedback introduced through the experimental configuration. This feedback mechanism stabilizes oscillations and is a common feature in control systems and nonlinear oscillators.

3. The transitions from limit cycles to quasi-periodic behavior, chaos, and back to limit cycles are presented as evidence of complex time crystal dynamics. However, these transitions are well-understood in the context of synchronization theory (see i.e. A. Pikovsky et al., Synchronization. A universal concept in nonlinear sciences (Cambridge University Press, Cambridge, 2010).):

- o Limit cycles correspond to synchronized oscillations where the system locks onto a dominant frequency.
- o Quasi-periodic behavior and chaos arise when synchronization breaks down due to mismatched frequencies or changes in feedback strength.

The reverse phase transition from chaos to limit cycles can also be explained through resynchronization phenomena, common in coupled oscillator systems (see Pikovsky again and A. Greilich et al., arXiv:2406.06243v1 as an example).

4. As one can see, there is currently a tendency to name a many-body interacting system with nonlinear feedback mechanism, that leads to self-oscillations, a time crystal. The paper relies on feedback-induced oscillations which are related to self-oscillatory systems with parametric excitation. In such systems, the oscillations themselves control changes in system parameters through feedback, leading to complex dynamical behavior. An example of this is a swing, where the timing of the pumping force induces sustained oscillations. Therefore, I would make a careful introduction to your system and put it in perspective with previous studies on self-oscillatory systems.

5. The current paper draws heavily on established principles from feedback-stabilized spin dynamics and nonlinear systems theory, as explored in the previous papers (Refs. [35,42,43]). The self-sustained oscillations, robustness due to feedback, and phase transitions between limit cycles, quasi-periodicity, and chaos are all phenomena well-documented in the spin maser literature and nonlinear dynamics. However, the framing of these behaviors as evidence of time crystals is the primary distinguishing factor of this paper. While this provides a novel perspective, it may overstate the originality of the findings, as the underlying physical mechanisms are consistent with those in previous studies.

6. While the paper employs chaos tests and the maximal Lyapunov exponent to characterize the system's behavior, it lacks analysis of the spatial complexity of the attractors. Calculating the correlation dimension would provide deeper insights into the system's dynamics. This additional metric would strengthen the claims regarding the nature of the observed dynamical phases.

Technical Questions and Clarifications:

7. Missing Methods Section:

The manuscript lacks details on critical experimental parameters, which are essential for the reproducibility and understanding of the results:

- What are the pressures and temperatures of the Rb and Xe gases?
- What is the fractional ratio of the two constituents, and how does this influence Rb-Xe interactions?
- How does the oscillation frequency depend on parameters such as magnetic field strength, spin-exchange interactions, feedback strength, and field gradients?

While some of this information may be available in related works (e.g., arXiv:2302.05264), it should be explicitly included in the current manuscript for completeness.

8. Optical Detection Setup:

The combination of the $\lambda/4$ plate, PEM, and linear polarizer (LP) in the detection scheme requires further clarification:

- How is the PEM aligned relative to the fast axis of the $\lambda/4$ plate?
- What retardation is used for the PEM—is it also set to $\lambda/4$?
- Does this setup mean you are primarily measuring the Faraday ellipticity signal rather than Faraday rotation? Why was this configuration chosen?
- I think a balanced photodiode setup would improve polarimetric sensitivity. Why was it not implemented?

9. Exploration of Alternative Feedback Configurations:

Have the authors explored other types of feedback configurations?

- For example, using the pump power or polarization instead of the magnetic field feedback.
- Applying feedback along different magnetic field directions—would this affect the dynamical phases observed?

10. Power Dependence of the Pump and Probe Beams:

- What are the powers (or power densities) of the pump and probe beams?
- Is there any dependence of the oscillation behavior on the beam power? This could influence the spin polarization and feedback strength, affecting the system dynamics.

The paper contributes to the ongoing exploration of nonlinear spin dynamics and precision measurement technologies, and its framing within the time crystal paradigm may stimulate further interdisciplinary research. However, the claims regarding the discovery of new phases of matter should be tempered with a deeper acknowledgment of the classical origins of the observed behaviors. I recommend major revisions to address the concerns outlined above. Specifically, the interpretation of the results should be reconsidered in light of established knowledge from nonlinear dynamics and feedback-stabilized spin systems. Additionally, more technical details and systematic investigations of parameter variations would strengthen the paper. At this point it is hard for me to accept the paper for a publication in Nature Communications. It is too similar to the previous research, partially by the same authors, see Refs. [35, 43], and requires more than just a rebranding of the observed oscillations. I will reassess the work upon the authors' response to these points and the feedback from other reviewers.

Reviewer #4

(Remarks to the Author)

The manuscript by Huang and collaborators reports on the experimental observation of continuous time crystals and continuous time quasi-crystals in a gas of spin particles (Xe, Rb). Using a feedback loop, namely acting on the system with a field whose strength depends on the system itself, the authors create a nonlinearity that can lead to a rich phase diagram. As one could expect from nonlinear ODEs of three variables (Eq. 1), tuning the parameters can lead to (i) limit cycles, (ii) oscillations at multiple incommensurate modes, and (iii) chaos. The authors argue for continuous time translational symmetry breaking, which elevates (i) to the status of continuous time crystal (CTC) and (ii) to that of continuous time quasi-crystal (CTQC). Robustness to noise is checked, and potential applications for metrology are argued for.

I find the paper interesting, relatively well written, and with the potential to have an impact in the field of time crystals. Before making a recommendation towards publication in Nature Communications, however, I would ask the authors to address a

key point on what elevates a limit cycle to a CTC, together with a couple of other minor comments.

Following [Science 377.6606, 670-673 (2022)], the authors argue that the key ingredient that elevates a limit cycle to the CTC state is that, upon repeating the experiment multiple times under identical conditions, the phase of the oscillations is different for every run, with a uniform distribution between 0 and 2π . In the experiment, a key action is the activation of the feedback. If the activation time is random, so will be the phase of the oscillations, but such randomness of the phase would follow from that of the feedback activation and thus not be indicative of CTC. The question is thus: can the feedback be activated quickly and precisely (with respect to the period of the CTC), so that it can be excluded as an obvious reason for the randomness of the CTC phase? A couple of numbers to quantify “quickly” and “precisely” in the previous sentence would be of help.

Related to the previous point. In the end, of course in the experiments there will be many sources of noise, and these can be amplified to ultimately determine the phase of the CTC. Yet, I like the analysis in Supplemental Section I, where the authors argue, even just as a matter of principle, that the randomness of the phases can result from the quantum fluctuations alone. The plot in Fig. 1 is good because it connects to the figures in the main, as well as to the figures in [Science 377.6606, 670-673 (2022)], where the concept of CTC was established. Yet, it would be of help to see some time traces. Presumably the time traces will show that oscillations are developed dynamically, and that the phase of these oscillations changes from run to run.

Random phase with respect to the phase of the feedback switch, or to the drive phase. In particular, how fast is that compared to the drive frequency?

The simulation in Supplementary Material is particularly interesting. I would like to see various time traces, the transient buildup of oscillations, the difference in phases from run to run.

Further minor comments

1. The authors mention a couple of experimental works on discrete time quasi-crystals. It could be worth mentioning the theory works, too. In particular, the idea that a magnetic field gradient can induce a second frequency, that is gradient-dependent and incommensurate, leading to quasi-crystallinity, was introduced for the Floquet case in [PRL 123.15, 150601 (2019)].

2. This is more of a stylistic matter, but I find the text bombastic and empty at times. For instance, I find the sentence in the abstract pretty empty: “However, their complex properties still require further exploration, and many phenomena remain to be revealed”. Adjectives such as “exciting”, “new”, “novel”, and “significant” could be toned down.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have addressed all my concerns and comments thus I am happy to recommend the work for publication in Nature Communications.

Reviewer #2

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Reviewer #3

(Remarks to the Author)

I have carefully read the authors' replies to my report and those of the other referees. I find the responses satisfactory and appreciate the substantial revisions and clarifications added to the manuscript, especially regarding the dynamical regimes, phase diagrams, and the connections to synchronization theory and nonlinear dynamics.

My main remaining comment is that the authors should be more careful in assigning the continuous time crystal (CTC) label based solely on the observation that the phase of oscillations covers the entire circle in repeated measurements. As demonstrated in [Kongkhambut, P. et al., Rep. Prog. Phys. 87, 080502 (2024)], when a system is subject to additional periodic driving, the phase distribution can become non-uniform or concentrated on particular sectors of the circle, rather than being randomly and uniformly distributed. In the present work, the external feedback used by the authors could, in certain parameter regimes, act similarly to periodic driving, potentially leading to a non-uniform phase distribution. Therefore, I would encourage the authors to avoid identifying the CTC phase solely on the basis of full phase circle coverage, and to discuss this subtlety in their manuscript, perhaps referencing the above work for comparison.

Altogether, I find the manuscript to be of high quality, and with the above caveat, I recommend it for publication.

Reviewer #4

(Remarks to the Author)

First of all, I would like to thank the authors for engaging in depth with the reports from the reviewers. My assessment of the paper in my first report was positive overall. In reference to the claim of the continuous time crystal (CTC) developing a random oscillation phase, which the authors argue as the distinctive feature of a CTC, I asked the authors to clarify the role played by the feedback activation time, and specifically whether this is short as compared to the CTC period. In their replies, the authors convincingly argue that this is indeed the case. Furthermore, the authors have added to the supplementary information a new Fig S2 showing the time traces of the dynamics, as I requested.

With respect to the other referee reports, I find that the comments from Reviewer 1 have been addressed satisfactorily. The most critical report is arguably that from Reviewer 3. I very much hear their point, namely that CTCs appear as a rebranding of known phenomena in nonlinear dynamical systems. The authors' reply hinges on the claim that the new and distinctive feature of CTCs is breaking of time translational symmetry, defined as in [Science 377.6606, 670-673 (2022)], namely in reference to the randomness of the phases of the oscillations. I find this ok, but it is however not textbook knowledge, and should be accordingly emphasized and argued for (for instance, it is currently not mentioned in the discussion, which I invite the authors to do).

Leaving aside the last point, that I leave to Reviewer 3's judgement, I overall find that the work meets the standards of Nature Communications and could be published there.

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Reply to Reviewer #1

Comment 1-1

In this work, Huang and colleagues report the observation of continuous time crystal (CTC) and quasi-crystal (CTQC) phases in a noble-gas nuclear spin system (Xe-129) coupled to Rb-87 spins. By introducing a gradient field and feedback coupling between the Rb and Xe nuclear spins, the system exhibits several dynamical phases: limit cycle, quasi-periodicity, and chaos, with additional signatures of spontaneous time-translation symmetry breaking. The manuscript also discusses the “melting” of these time-ordered phases into chaotic motion and their eventual re-formation at stronger gradient fields.

In my opinion, the topic is timely, echoing with the growing interest in nonequilibrium phases of matter and time-crystal behavior. The experimental approaches using spin-exchange collisions, feedback loops, and gradient fields provide a novel platform to investigate such non-equilibrium many-body phenomena and could potentially motivate further exploration. However, I am not fully convinced by several key aspects in the current manuscript. So I would suggest the authors to address the following comments before I can make a recommendation for the publication at Nature Communications.

Our response: We would like to thank the reviewer for his/her careful reading and potential recommendation of our work. The insightful and pertinent comments and suggestions help us to greatly improve the quality of our manuscript. On the basis of these comments, we have carefully revised our main text and Supplementary Information. For example, we have conducted new experiments to obtain a phase diagram of the time-crystal system. Additionally, we have incorporated new sentences in the main text to facilitate a more intuitive understanding of the intriguing time-crystal behaviors observed in our system, such as the emergence of quasi-periodic oscillations and the “re-emergence” of time-crystalline order from chaos. Moreover, we have added a new section in the Supporting Information, providing a detailed derivation of the nonlinear Bloch equation from the underlying spin Hamiltonian, along with intuitive explanations in the main text. Below, we provide the detailed reply to the comments and the corresponding major changes in the revised paper.

Comment 1-2

Major Points

1. The derivation of equation (1) and underlying Hamiltonian: Equation (1) (the set of nonlinear Bloch equations) is the central part of the paper’s argument. However the authors provide limited detail on how it is derived from a the Hamiltonian governing the system interaction. It is not immediately clear how interactions, pumping, feedback, and approximations enter into these effective equations. I would recommend the author to provide a detailed derivation by starting from the relevant spin Hamiltonian in a magnetic field. This derivation can go into the Methods/Supplementary Materials. However in the main text, it would be good to provide a high-level intuition on how to obtain the Eqn.1 from the full quantum Hamiltonian.

Our response: We would like to thank the reviewer for pointing out the lack of description. As pointed out by the reviewer, the dynamics of noble-gas atoms in our work are indeed governed by the underlying spin Hamiltonian, from which Equation (1) can be derived. To provide a clearer explanation, we first identify the relevant spin Hamiltonian and its connection to Equation (1),

and then provide a detailed derivation of each spin Hamiltonian.

The spin Hamiltonian for the noble-gas system is expressed as

$$H = -\mu_K \cdot \mathbf{B}_z - \mu_K \cdot \mathbf{B}_f + \alpha \mathbf{S} \cdot \mathbf{K}, \quad (1)$$

where the first two terms represent the Zeeman interaction of the nuclear spin with the applied static magnetic field B_z and the the feedback magnetic field $\mathbf{B}_f$, and the last term describes the spin-exchange interaction between the noble-gas nuclear spin $\mathbf{K}$ and the electron spin $\mathbf{S}$ of the alkali-metal atoms. The nuclear magnetic moment is given by $\mu_K = \gamma \mathbf{K}$, where γ denotes the gyromagnetic ratio of noble gas. Each term in Equation (1) corresponds to a specific contribution from the Hamiltonian. The explicit form of Equation (1) is provided below, with the related spin Hamiltonian labeled accordingly

$$\begin{cases} \dot{P}_x(\omega) = \overbrace{\omega P_y(\omega)}^{-\mu_K \cdot \mathbf{B}_z} + \overbrace{\alpha P_z(\omega)}^{-\mu_K \cdot \mathbf{B}_f} \int \rho(\omega') P_x(\omega') d\omega' - \overbrace{\frac{P_x(\omega)}{T_2}}^{\text{relaxation}} \\ \dot{P}_y(\omega) = -\overbrace{\omega P_x(\omega)}^{-\mu_K \cdot \mathbf{B}_z} - \overbrace{\frac{P_y(\omega)}{T_2}}^{\text{relaxation}} \\ \dot{P}_z(\omega) = \overbrace{\alpha P_x(\omega)}^{-\mu_K \cdot \mathbf{B}_f} \int \rho(\omega') P_x(\omega') d\omega' - \overbrace{\frac{P_z(\omega)}{T_1}}^{\text{relaxation}} + \overbrace{R_{SE}}^{\alpha \mathbf{S} \cdot \mathbf{K}} \end{cases}$$

In the following analysis, we present a comprehensive derivation of Equation (1) originating from the associated spin Hamiltonian, carefully detailing the approximations applied throughout.

(1) Spin-exchange interactions. Noble gas lacks any optical transition from the ground levels, making them difficult to polarize directly using optical pumping methods. Instead, noble gases can be polarized through spin exchange when placed in the same vapor cell as optically pumped alkali-metal atoms. Spin polarization is transferred between these alkali-metal atoms and the noble gas nuclei during binary collisions and through the formation and destruction of van der Waals molecules in three-body collisions. The magnetic dipole hyperfine interaction is the primary mechanism responsible for this spin exchange, described by the Hamiltonian:

$H_{SE} = \alpha \mathbf{S} \cdot \mathbf{K}$. The coupling coefficient is given by $\alpha = \frac{8\pi g_S \mu_B \gamma}{3} |\psi(0)|^2$, where $|\psi(0)|^2$ represents the probability density of the alkali-metal atom's valence electron at the site of the noble-gas nucleus, g_S denotes the electron's Landé g-factor, and μ_B is the Bohr magneton. By averaging over all types of collisions, the rate of change of the noble gas's density matrix ρ due to spin-exchange interaction is expressed as follows [see, for example, S. Applet et al., Phys. Rev. A 58, 1412 (1998)]:

$$\begin{aligned} \frac{d}{dt} \rho = & (\Gamma_{bc} + \Gamma_{vdw,S}) \sum_{m,n=x,y,z} 4 \langle S_m S_n \rangle \left(K_n \rho K_m - \frac{1}{2} \{K_m K_n, \rho\} \right) \\ & + \Gamma_{vdw,F} \sum_{m,n=x,y,z} 4 \langle F_m F_n \rangle \left(K_n \rho K_m - \frac{1}{2} \{K_m K_n, \rho\} \right) + \frac{1}{i} [\mathcal{E}_{SE}, \rho], \end{aligned} \quad (2)$$

where Γ_{bc} , $\Gamma_{vdw,S}$ and $\Gamma_{vdw,F}$ denote the binary collision rate and the three-body formation rates of molecules with “very short” and “short” lifetimes, respectively; and F represents the total angular

momentum of the ^{87}Rb atoms. The frequency-shift operator $\mathcal{E}_{\text{SE}} = \frac{8\pi g_S \mu_B \gamma}{3} \kappa [\text{Rb}] \langle \mathbf{S} \rangle \cdot \mathbf{K}$ can be understood as the effect of an effective magnetic field $\mathbf{B}_{\text{SE}}$ generated by the spin-polarized alkali-metal atoms with a number density $[\text{Rb}]$. We define the spin polarization as $\mathbf{P} = \langle \mathbf{K} \rangle / K = 2 \langle \mathbf{K} \rangle$. By multiplying both sides of the equation by $2\mathbf{K}$ and taking the trace, we find that the evolution of the nuclear spin polarization is given by

$$\begin{aligned} \frac{d}{dt} \mathbf{P} &= (\Gamma_{\text{bc}} + \Gamma_{\text{vdw,S}})(2\langle \mathbf{S} \rangle - \mathbf{P}) + \Gamma_{\text{vdw,F}}(2\langle \mathbf{F} \rangle - \langle 2\mathbf{F} \cdot \mathbf{F} - \mathbf{F}\mathbf{F} \cdot - (\mathbf{F}\mathbf{F})^\dagger \cdot \rangle \mathbf{P}) - \gamma \mathbf{B}_{\text{SE}} \times \mathbf{P} \\ &= 2 \left[(\Gamma_{\text{bc}} + \Gamma_{\text{vdw,S}}) \langle \mathbf{S} \rangle + \Gamma_{\text{vdw,F}} \langle \mathbf{F} \rangle \right] - \left[\Gamma_{\text{bc}} + \Gamma_{\text{vdw,S}} + \Gamma_{\text{vdw,F}} \langle 2\mathbf{F} \cdot \mathbf{F} - \mathbf{F}\mathbf{F} \cdot - (\mathbf{F}\mathbf{F})^\dagger \cdot \rangle \right] \mathbf{P} \\ &\quad - \gamma \mathbf{B}_{\text{SE}} \times \mathbf{P}. \end{aligned} \quad (3)$$

With the presence of a continuous pump laser along the z axis, the spin polarization of the alkali-metal atoms can be approximated as a constant $\langle \mathbf{S} \rangle = S_0 \hat{z}$. In this case, the ^{87}Rb states that can be characterized by spin-temperature distribution, where $\langle F_z \rangle / \langle S_z \rangle = 2 \langle \mathbf{F} \cdot \mathbf{F} - F_z^2 \rangle = q$. Assuming the effective field is negligible compared to the bias and feedback fields, the evolution equation for the nuclear spin polarization due to spin-exchange collisions reduces to

$$\frac{d}{dt} P_z = -\Gamma_{\text{SE}} P_z + R_{\text{SE}}, \quad (4)$$

where $\Gamma_{\text{SE}} = \Gamma_{\text{bc}} + \Gamma_{\text{vdw,S}} + q\Gamma_{\text{vdw,F}}$. The first term $-\Gamma_{\text{SE}} P_z$ contributes to the relaxation, and $R_{\text{SE}} = 2\Gamma_{\text{SE}} S_0$ represents the noble-gas spin pumping originating from the spin-exchange interaction between alkali-metal and noble-gas spins.

(2) Feedback interaction. According to our measurement-feedback configuration, the average noble-gas spin polarization $\bar{P}_x$ is measured and subsequently fed back as a magnetic field denoted by $\mathbf{B}_f = -\alpha \bar{P}_x / \gamma \hat{y}$. Here α represents the feedback strength, which can be experimentally calibrated. Analogous to the externally applied magnetic field, the feedback magnetic field interacts with the nuclear spin via the Zeeman interaction. The specifics of this Zeeman interaction are elaborated upon in the following.

(3) Zeeman interaction. The Zeeman interaction causes the density operator of the noble-gas nuclear spin to evolve as follows

$$\frac{d}{dt} \rho = \frac{1}{i} [-\gamma \mathbf{K} \cdot (\mathbf{B}_z + \mathbf{B}_f), \rho]. \quad (5)$$

By multiplying both sides by $2\mathbf{K}$ and taking the trace, we derive the time evolution of the nuclear spin polarization

$$\frac{d}{dt} \mathbf{P} = \gamma \mathbf{P} \times (\mathbf{B}_z + \mathbf{B}_f). \quad (6)$$

Substituting $\mathbf{B}_z = B_z(\mathbf{r}) \hat{z}$ and $\mathbf{B}_f = -\alpha \bar{P}_x / \gamma \hat{y}$ into this equation yields the component evolution equations

$$\begin{cases} \dot{P}_x = \omega P_y + \alpha P_z \bar{P}_x \\ \dot{P}_y = -\omega P_x \\ \dot{P}_z = \alpha P_x \bar{P}_x \end{cases} \quad (7)$$

where $\omega = \gamma B_z(\mathbf{r})$. Given that ω is a function of position $\mathbf{r}$, the polarization must also be treated as a function of position, or more meaningfully, as a function of the Larmor frequency ω . Thus, we express $P_i = P_i(\omega)$ for $i = x, y, z$. The averaged polarization component $\bar{P}_x$ can be written

as $\bar{P}_x = \int \rho(\omega') P_x(\omega') d\omega'$. Incorporating $P_i = P_i(\omega)$ and $\bar{P}_x$ into the component evolution equations allows us to derive the corresponding parts in Equation (1) of the main text.

(4) Spin relaxation. The polarization of noble-gas nuclear spins can be disrupted or altered by various complex relaxation mechanisms. Notably, these include collisions with alkali-metal and other noble-gas atoms, spatial diffusion between regions of differing spin polarization, and spin relaxation during adsorption on vapor-cell wall. There are many works on these relaxation mechanisms, such as the comprehensive study by Happer et al., "Optically Pumped Atoms" (John Wiley & Sons, 2010). Ultimately, these mechanisms can be described semiclassically in terms of relaxation times T_1 and T_2 , which are incorporated into the evolution equations.

Considering this, we have added a new section titled "**Model and detailed derivations**" in Supplementary Information. The supplementary content is nearly the same with that presented above. In addition, we have added several new sentences in the revised main text to clarify the origin of each term in Equation (1), for example,

on page 2, "*Under this feedback scheme, the system dynamics are governed by a Hamiltonian that includes Zeeman interactions with the gradient and feedback magnetic field, spin-exchange interactions with polarized ^{87}Rb atoms, and relaxation mechanisms. Further details can be found in the Supplementary Information*".

Comment 1-3

2. Related to 1, my current understanding is that the nonlinear Bloch equations presented resemble a classical spin system rather than a quantum system, i.e. $P(w)$ in eqn.1 is a number, rather than operator. Is this understanding correct? If so, could the authors elaborate on this a bit more. Seems that recent studies have shown that time crystal behavior should also persist in classical spin systems.

Our response: We would like to thank the reviewer for pointing out this insightful comment. As discussed in **Comment 1-2**, our system is a quantum spin system governed by a series of spin-related Hamiltonians. Despite this, it is important to note that the quantum fluctuations of our spin system are negligible compared to our current detection sensitivity, allowing us to disregard these fluctuations. In this situation, the dynamics of our spin system can be effectively described by the evolution of the expectation values of spin operators, such as $\langle \mathbf{K} \rangle$. This approximation significantly simplifies the dynamic equations. Furthermore, because our system is a spin-1/2 system, the evolution of the operator expectation value can be formally represented by the classical Bloch equations.

Considering this, we have added several new sentences in the main text to to clarify it, for example,

on page 2, "*Neglecting quantum fluctuations, the dynamics of ^{129}Xe spins can be described by the evolution of spin operator expectation values. Here the spin polarization is defined as the normalized expectation value of ^{129}Xe spin operator, $\mathbf{P} = 2\langle \mathbf{K} \rangle$. For our spin-1/2 system, the evolution of the polarization can be formally expressed as a set of nonlinear Bloch equations*".

Comment 1-4

3. Here the main novelty is that the authors observe quasi-periodic oscillations with multiple incommensurate frequencies, i.e. continuous time quasicrystal. However, it is not clear at the

moment what key ingredients in their experimental system allow the observation of this quasi-crystal behavior. In particular, compared to the cited works on CTC using other experimental platforms, could the author discuss more explicitly what are the unique ingredients (e.g., the role of the gradient field, feedback strength, etc.) that enable this quasi-crystal behavior?

Our response: We thank the reviewer for highlighting the novelty of our observation of quasi-periodic oscillations with multiple incommensurate frequencies, indicative of continuous time quasicrystal (CTQC) behavior. The key ingredient in our system is the controllable magnetic field gradient. Below, we would like to provide some intuition to enhance understanding of this.

A linear magnetic gradient applied along the z -bias field introduces a spatially dependent detuning across the spin ensemble, resulting in a broad distribution of Larmor frequencies ω_j . This configuration, when combined with feedback, gives rise to effective nonlinear coupling among spin modes with different ω_j , enabling complex dynamics. In this context, continuous time crystals (CTCs) can be interpreted as a manifestation of synchronization among these modes, where spins with different ω_j collectively oscillate at a common frequency. However, as the detuning or the strength of the nonlinear coupling is varied, this frequency synchronization can break down, transitioning the system into a quasi-periodic or even chaotic regime. The number of distinct frequencies ω_j induced by the magnetic gradient is, in principle, very large, making comprehensive analytical treatment challenging. To gain insight into the underlying physical mechanism, we analyze a simplified model comprising two dominant spin modes with frequencies ω_1 and ω_2 . When the detuning $\epsilon = \omega_1 - \omega_2$ is small, nonlinear coupling readily synchronizes the two modes, resulting in a stable limit cycle. As ϵ increases, synchronization breaks down via a Hopf bifurcation, characterized by a pair of unstable limit cycles colliding with a pair of stable fixed points [S. H. Strogatz, Nonlinear dynamics and chaos: with applications to physics, biology, chemistry, and engineering, CRC press (2018)]. Beyond this bifurcation, the system enters quasi-periodic or chaotic regimes. In the regime of large detuning, the nonlinear coupling is no longer sufficient to maintain full frequency synchronization. Instead, it mixes ω_1 and ω_2 into a composite response, giving rise to quasi-periodic behaviors. When extended to many spin modes, the system can similarly undergo multiple bifurcations and exhibit increasingly complex dynamical phenomena.

Due to the universal nature of nonlinear synchronization, quasi-crystal behavior can, in principle, be realized across a variety of platforms by combining magnetic field gradients with feedback. This includes systems such as alkali-metal atomic gases, nuclear-spin liquids, and others.

Considering this, we have added several new sentences to the “**Discussion**” section of the main text. These additions emphasize the role of the magnetic-field gradient as a crucial component and explore potential extensions of our approach to other platforms capable of realizing quasi-crystal behavior. For example,

on page 6 in the main text, “We introduce a key ingredient—a magnetic field gradient—that enriches the dynamical behavior of the noble-gas spin system, enabling the emergence of quasi-periodic and chaotic phenomena. While our demonstration focuses on a Rb-Xe hybrid gas system, we believe that integrating magnetic field gradients with feedback, as demonstrated in our work, is widely applicable across various platforms. This includes systems such as alkali-metal atomic gases, nuclear-spin liquids [M. Suefke et al., Nat. Phys. 13, 568 (2017)], and others”.

Comment 1-5

4. Time-Window Analysis: The authors mention that time-ordered phases (CTC or CTQC) only persist over a certain “stationary window,” which becomes narrower if one measures longer (since this is a continuous situation, one does not need to worry about the lifetime of the spins). The “melting” into chaos is super interesting, but it is not clear how to distinguish in a robust way between the different dynamical regimes (e.g. for Fig. 2h and 2i, without the lines, it is not immediately obvious to me the difference. Could the authors provide additional experimental with a few varying time windows to show how the system transitions from time-ordered oscillations to chaotic dynamics.

Our response: We would like to thank the reviewer for pointing out this insightful comment. To clearly illustrate the transition process between different time-ordered phases, we aim to provide additional details from two perspectives to help prevent potential misunderstandings.

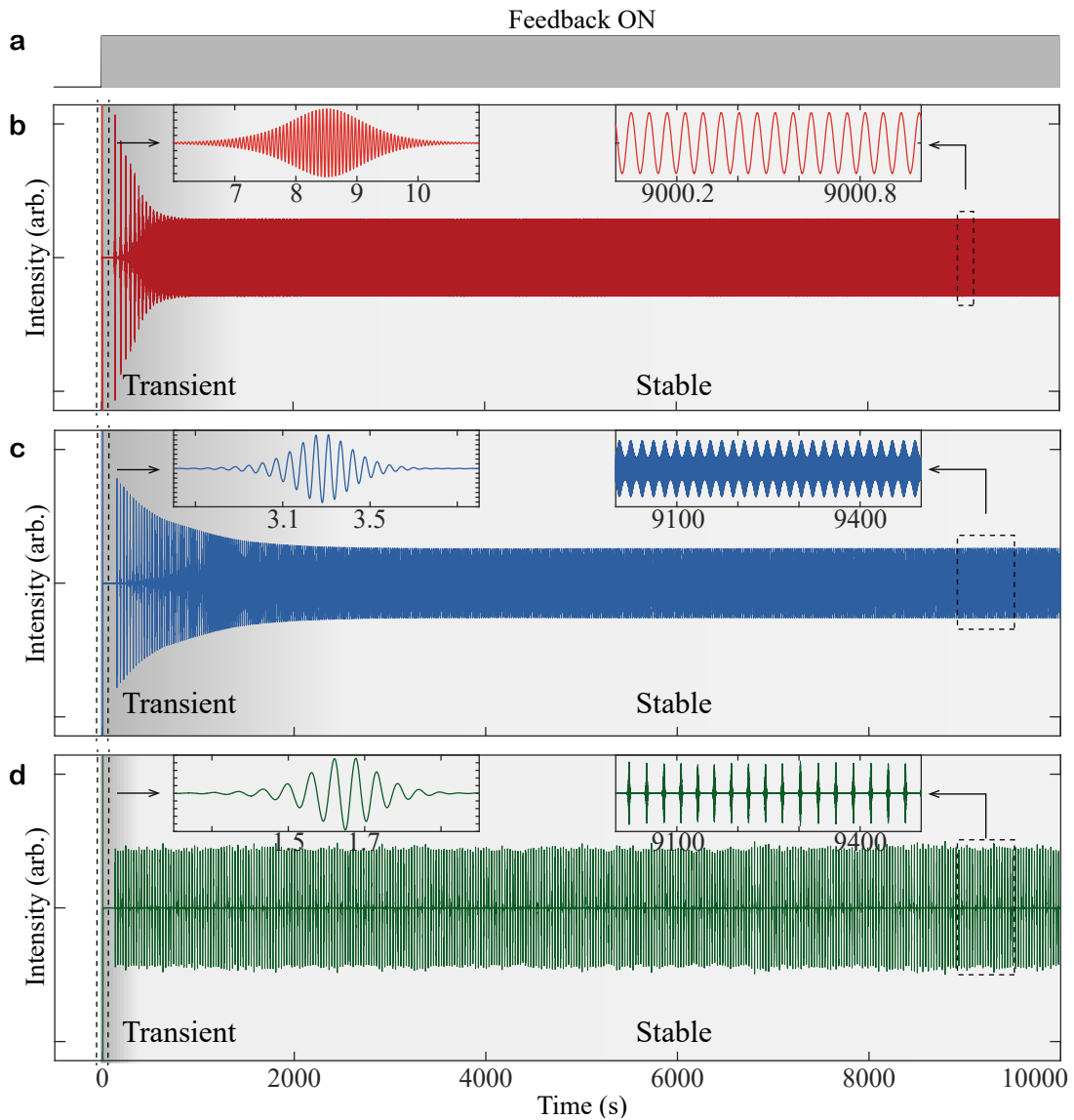


Figure 1: **Establishment of elementary oscillation pattern.** **a**, Timing sequence of the feedback switch. Representative time traces showing the transient build-up and stabilization of oscillations in the limit cycle phase (**b**), quasi-periodic phase (**c**), and chaotic phase (**d**). The time traces presented in Fig. 2d, e, and f of the main text are extracted from an arbitrary 1000-s time window within the stationary state.

(1) It is important to note that the system does not transition between dynamical phases within a single experimental run over varying time windows. Instead, the transitions occur across distinct experiments where system parameters are varied. Specifically, the three dynamical phases illustrated in Fig. 1 are measured in three separate experiments, each under different parameter settings. In each run, with a fixed parameter set, the dynamical phase remains stable. For example, in Fig. 1b, once the limit cycle state is established, it persists continuously.

On the other hand, when system parameters are altered, the dynamical phase may change. In the main text, Figure 3 provides further illustration of how the system transitions between time-ordered phases and a chaotic phase as a function of varying parameters, including feedback strength and magnetic-field gradient. In particular, in response to the suggestions made in **Comment 1-6**, we have designed and conducted new experiments to present a two-dimensional phase diagram. This diagram offers a more detailed depiction of the transition process from one dynamical phase to another under varying parameters.

(2) As pointed out by the reviewer, distinguishing between time-ordered phases and the chaotic phase is crucial, and relying solely on time-domain signals or frequency-domain spectral lines is not a sufficiently robust method. In our work, we employ the Lyapunov Exponent, a widely used tool for characterizing chaos, which quantifies the exponential rate of divergence of nearby trajectories in reconstructed phase space. A positive largest Lyapunov Exponent indicates chaos. Additionally, we use the Chaos Decision Tree Algorithm, a method recently applied in works such as [Nat. Phys. 20, 631 (2024)]. This algorithm processes time series data and outputs a chaos criterion value, K , where values approaching 1 signify chaotic systems, while values near 0 indicate periodic or quasi-periodic systems. Our results show strong agreement between the K -value and Lyapunov Exponent, confirming the robustness of our approach.

Considering this, we have added a new section titled “**Phase Diagram over a Two-Dimensional Parameter Span**” in Supplementary Information and added the corresponding new sentences in the main text. More detailed information can be found in **Comment 1-6**. Furthermore, we provide additional experimental results with longer observation time. Based on this, we have made several revisions to the manuscript,

on page 3 in the main text, “*In the experiment, after a transient build-up process, the system levels into stable oscillations and maintains stable dynamic behaviors. The details of the establishment process are provided in the Supplementary Information. For demonstration, we extract 1000-second segments from the steady states of the three distinct dynamical phases. Figure 2d shows a representative time trace along with the corresponding spectrum for the limit cycle phase. The time trace reveals impeccably periodic oscillations, showcasing both a stable amplitude and frequency. By performing ...*”

In the Supplementary Information, we have added a new figure and corresponding descriptions,

“... we begin by waiting approximately 2000 seconds with the feedback circuit disconnected to ensure that the ^{129}Xe spin polarization is fully aligned along the z axis. Once the feedback is activated, the system undergoes a transient phase before settling into stable oscillations, exhibiting distinct dynamic behaviors. Figure S4 illustrates the timing sequence of the feedback switch, along with representative time traces showing the establishment and stabilization of oscillations in the three dynamical phases”.

5. The feedback strength α and the field gradient g are core tunable parameters in the experiments. While Fig 3 and 4 shows the effect of tuning both α and g , it would be helpful if the authors could provide a more systematic exploration of the (α, g) parameter space to show a phase diagram-like plot of the CTC CTQC behaviors.

Our response: We would like to thank the reviewer for this very valuable suggestion, which really helps us improve the quality and clarity of our manuscript. In order to more comprehensively demonstrate the dependence of the system's dynamical phase on the feedback strength α and the field gradient g , we have designed and performed a new set of experiments to systematically explore the (α, g) parameter space. The resulting phase diagram is presented in Fig. 2a (see below), and the K values, which are used to distinguish the dynamical phases, are calculated and shown in Fig. 2b (see below). Below, we detail the experimental procedure used to obtain this phase diagram and discuss the new insights gained from it.

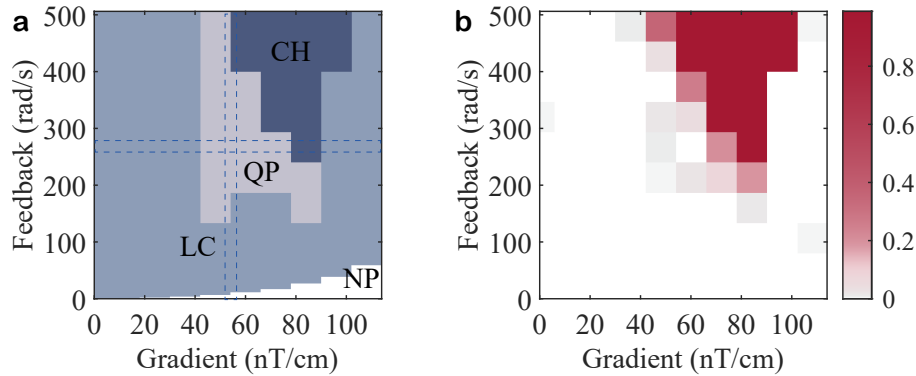


Figure 2: Experimental phase diagram spanned by (α, g) showing distinct regimes (a): the (NP) no signal point, (LC) limit cycle, (QP) quasi-periodic orbits, and (CH) chaos; and corresponding K value (b).

(1) We first detail the newly conducted phase-diagram experiments. To construct the two-dimensional phase diagram, we maintain a constant magnetic-field gradient g while varying the resistance in series with the feedback-field coil to adjust the feedback strength α . This process is repeated for various g values to scan the parameter space. To thoroughly explore the parameter space, we begin by waiting approximately 2000 seconds with the feedback circuit disconnected to ensure that the ^{129}Xe spin polarization is fully aligned along the z axis. Once the feedback is activated, the system undergoes a transient phase before settling into stable oscillations, exhibiting distinct dynamic behaviors. We measure the intensity changes of the probe beam, which is directly proportional to the average ^{129}Xe spin polarization, after the system dynamics have stabilized. For example, in the chaotic regime, data collection extends up to 10,000 seconds after the feedback is activated. To differentiate between chaotic and time-ordered dynamics (such as limit cycles and quasi-periodic behavior), we utilize the Chaos Decision Tree Algorithm. This method analyzes time series data and outputs a parameter K ; values approaching 1 indicate chaos, while values near 0 suggest periodic or quasi-periodic dynamics. Limit cycles and quasi-periodic behaviors can be intuitively distinguished by their spectral characteristics: quasi-periodicity is indicated by multiple peaks with incommensurate frequencies, whereas limit cycles are represented by a single peak, possibly with harmonics, though such harmonics were not observed in our experiments.

(2) We now discuss the new information obtained from the phase diagram. Figure 2 (see above)

presents the experimental phase diagram spanned by (α, g) , showing distinct regimes: the no signal point, limit cycle, quasi-periodic orbits, and chaos. Compared to 1D parameter sweep of the previous version, the newly obtained 2D phase diagram offers a more comprehensive insight, in particular, we obtain some quantitative results. First, when the feedback strength is weak, ^{129}Xe spins exhibit a normal phase (no signal), with polarization aligned along z . As the feedback strength exceeds a critical threshold, the spin system enters a limit cycle phase. The threshold becomes higher as the magnetic field gradient increases. Second, only when the gradient exceeds a certain value (~ 40 nT/cm), phases beyond the limit cycle (such as quasi-periodic and chaotic dynamics) emerge, suggesting that the gradient plays a crucial role in the emergence of new dynamic phases. Third, after being supplemented by the 2D phase diagram, the conclusions drawn from the two phase transition curves in the previous version remains valid within a certain parameter range: over a broad gradient range (50 – 100 nT/cm), the system transitions from no signal to limit cycle, then to quasi-periodic phase, and finally to chaos as the feedback strength increases; over a broad feedback range ($\gtrsim 250$ rad/s), the system transitions from limit cycle to quasi-periodic phase, then to chaos, and eventually returns to limit cycle phase as the gradient increases. The portions enclosed by the dashed boxes in Fig. 2 correspond to the two phase transition curves of the previous version. Fourth, as the gradient increases, the quasi-periodic region becomes narrower, which may indicate an underlying competition mechanism within the system.

Considering this, we have added a new section titled “**Phase diagram over a two-dimensional parameter span**” in the Supplementary Information. The supplementary content is nearly the same with that presented above. In the main text, we have also added several new sentences to introduce this

on page 5 in the main text, “ *In order to more comprehensively demonstrate the dependence of the time crystal behavior on the feedback strength α and the field gradient g , a phase diagram spanning the two-dimensional parameter space (α, g) is provided in the Supplementary Information* ”.

Comment 1-7

Minor Comment

6. The manuscript briefly describes how the system transitions from a time-crystalline (CTC or CTQC) phase into a chaotic one and back again (“re-formation”). These abrupt transitions appear to be quite interesting to me. However, there are not too much discussions around it. Could the authors comment more on this, e.g. provide some intuition on such behavior, and when this will appear and how does this depend on α and g .

Our response: We appreciate the reviewer bringing this interesting aspect of our observations to our attention. The transition from time-crystalline to chaotic dynamics, and its subsequent “re-emergence”, is indeed a fascinating yet complex phenomenon, largely due to the system’s intrinsic nonlinearity. Inspired by the reviewer’s insightful comment, we have made a concerted effort to delve deeper into this behavior and have reviewed relevant literature to develop a more intuitive and visually guided understanding. Below, we summarize our intuitive understanding.

This “re-emergence” transition can be interpreted within the general framework of nonlinear dynamics in complex systems—particularly through the perspective of synchronization, for example, as illustrated in a recent study, Ref. [A. Greilich et al., Nat. Commun. 16, 2936 (2025)].

In general synchronization theory, oscillators with different natural frequencies can, through interactions, adjust their frequencies to achieve collective synchronization. In our system, each nuclear-spin mode with a distinct Larmor frequency ω can function as an individual oscillator. These modes are globally coupled through the nonlinear feedback field (with coupling strength α), while the applied magnetic field gradient (characterized by g) introduces a continuous detuning across the ensemble—analogue to frequency mismatch in classical synchronization models. At moderate values of both α and g , the feedback-mediated coupling overcomes the detuning, leading to collective synchronization or quasi-periodic behavior—manifested as continuous time crystals (CTC) or continuous time quasi-crystals (CTQC). However, as g increases while keeping α fixed, the detuning becomes too large, causing synchronization to break down and the system to enter a chaotic regime. Interestingly, at very large values of g , while the range of spin modes ω broadens, the number of spins associated with each specific mode decreases. This reduction favors coupling dominance once again, enabling the system to resynchronize and thereby leading to the observed re-formation of the CTC phase from chaos.

This above interpretation provides an intuitive explanation. However, a full analytical characterization of the phase diagram in the (α, g) parameter space still remains challenging due to the high dimensionality and complexity of the nonlinear interactions. Fortunately, it is possible to experimentally characterize the phase transition. In our revised manuscript, we have designed and performed new experiments to show how the dynamical phase evolve as a function of α and g —now illustrated in the newly added diagrams (Fig. 2) in the Supplementary Information.

Considering this, we have added several new sentences to provide some intuition on such complex behaviors,

on page 5 in the main text, “...re-emergence of a limit cycle phase above a critical value of $g \approx 92 \text{ nT/cm}$. These dynamical transitions can be understood as bifurcations or as the onset and breakdown of synchronization. For example, at large magnetic field gradients, the broader frequency distribution $\rho(\omega)$ decreases the number of spins contributing to each frequency mode, thereby facilitating resynchronization and enabling the re-emergence of a limit cycle from chaos. Similar transitions have been observed in other nonlinear systems—for instance, in the Lorenz system, increasing the Rayleigh parameter r beyond a critical threshold induces a transition from chaotic behavior back to a limit cycle [S.H. Strogatz, *Nonlinear Dynamics and Chaos*, Chapman and Hall/CRC, 2024; C. Sparrow, *The Lorenz Equations*, Springer, 1982]”.

Reply to Reviewer #2

Comment 2-1

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Our response: We really appreciate the reviewer's co-review. His/her very valuable comments help us to improve the quality of our manuscript. We believe that we have addressed well the reviewer's comments, and hope that the reviewer now find the paper suitable for Nature Communications.

Reply to Reviewer #3

Comment 3-1

The authors report on the experimental observation of Continuous Time Crystals (CTCs) and Continuous Time Quasi-Crystals (CTQCs) in noble-gas nuclear spins. Utilizing a system of ^{129}Xe spins coupled to optically pumped ^{87}Rb atoms, they introduce an external feedback mechanism that sustains persistent oscillations of nuclear spin polarization. The feedback induces nonlinear interactions between the spins, leading to rich dynamical phases, including:

- Limit cycle oscillations, identified as CTCs, characterized by persistent, periodic behavior with random time phases.
- Quasi-periodic oscillations, associated with CTQCs, where the system exhibits multi-frequency behavior akin to spatial quasi-crystals.
- Chaotic oscillations, interpreted as the "melting" of time crystals.

The paper highlights the robustness of these phases against noise perturbations and the observation of an unusual reverse phase transition from chaos back to limit cycles. The authors claim that these findings open new avenues in precision measurements, such as multimode masers and quantum metrology.

The manuscript is well-written and offers a fresh perspective by bridging concepts from nonlinear dynamics and condensed matter physics. However, several concerns arise regarding the interpretation of the results and the technical depth of the analysis.

Our response: We are very grateful to the reviewer for considering our manuscript "well-written and offers a fresh perspective". We also sincerely appreciate the reviewer's thoughtful comments and constructive suggestions, which significantly inspired us to improve the quality of our manuscript in many ways. Based on these valuable comments, we have made major revisions to both the main text and the Supplementary Information. In particular, we have clarified that although the random-phase behavior of self-oscillations arises naturally from the system's initial conditions and nonlinear dynamics, not all such self-oscillations exhibit random phase covering the range from 0 to 2π . This characteristic constitutes a distinctive signature of spontaneous symmetry breaking in time crystals. As raised by the reviewer regarding synchronization theory, we review the relevant literature and find that this framework offers a compelling explanation of the observed phenomena. Accordingly, we have made a more careful introduction to our system and result using the common framework established for the nonlinear systems. In addition, we have carefully expanded the Supplementary Information to include detailed descriptions of our spin system, spin-exchange optical pumping, the optical detection setup, and the feedback configuration, along with analysis of parameter dependence, thereby enhancing the technical depth. Below, we provide the detailed reply to each comment and the corresponding major changes in the revised manuscript.

Comment 3-2

1. The authors attribute the random phase of oscillations in repeated runs to spontaneous breaking of continuous time-translation symmetry, framing it as evidence of time crystal behavior. However, this random phase behavior is a well-known property of self-oscillating systems with feedback. It is a natural consequence of the system's initial conditions and nonlinear dynamics,

rather than a unique signature of a new phase of matter.

Our response: We thank the reviewer for highlighting this important aspect. Indeed, the random phase of oscillations is not a unique signature of time crystals and may arise naturally from the system's initial state and nonlinear dynamics. At the same time, it is important to note that not all self-oscillations can form continuous time crystals (CTCs), which require more stringent conditions. For a system to be classified as a CTC, it must exhibit robustness against temporal perturbations and be associated with the spontaneous breaking of continuous time translation symmetry.

When continuous time translation symmetry is spontaneously broken, the phase of self-oscillations for repeated realizations assumes random values across the entire interval from 0 to 2π , as demonstrated in Ref. [Science 377, 670-673 (2022)]. In contrast, the self-oscillations whose random phases are restricted to a subset of $[0, 2\pi]$ cannot be considered as CTCs. We have performed new numerical simulations to verify this point. In our case, for example, when parameters are set outside our experimental range — $\omega_c/2\pi = 0.012$ Hz with no magnetic field gradient ($g = 0$) and $\alpha = 1.92$ rad/s — 100 simulations initialized with fluctuation $\delta\bar{P}_i = 10^{-9}$ yield a phase distribution that fails to cover the full range of $[0, 2\pi]$, as shown in Fig. 3b. This represents a trivial limit cycle rather than a CTC, differing from the result of a CTC in Fig. 3a, where the phase distribution is nearly uniform. Thus, the 2π random phase distribution arising from the spontaneous breaking of continuous time translation symmetry is a critical signature of a CTC.

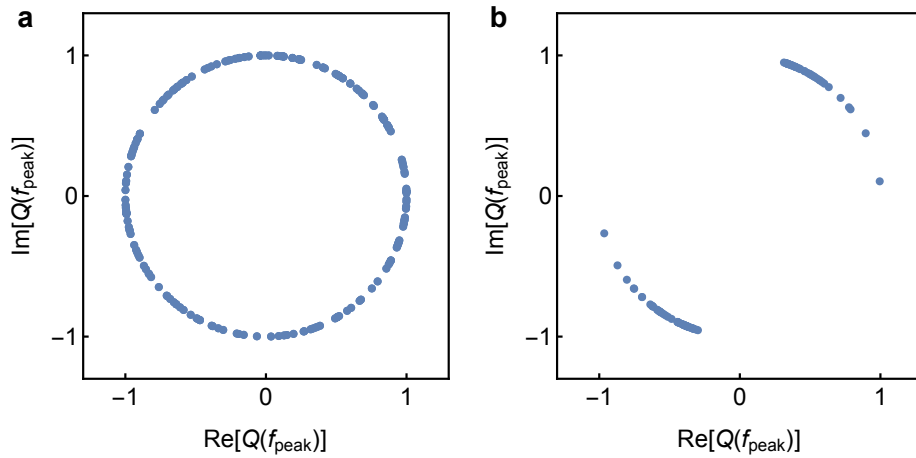


Figure 3: **a**, Phase distribution of a nontrivial limit cycle (a CTC), where we take $g = 0.065$ nT/cm, $\alpha = 76.92$ rad/s and $\omega_c/2\pi = 17.5$ Hz, and run 100 repeated simulations with initial fluctuation strength $\delta\bar{P}_i = 10^{-9}$. The data is distributed over the unit circle, indicating a phase distribution of 2π . **b**, Phase distribution of a trivial limit cycle (not a CTC), where we take $g = 0$, $\alpha = 1.92$ rad/s and $\omega_c/2\pi = 0.012$ Hz, and also run 100 repeated simulations with initial fluctuation strength $\delta\bar{P}_i = 10^{-9}$. But the data is distributed on two discontinuous arcs, which do not meet the 2π random phase required by the CTC.

Considering this, we have added a new example of self-oscillations whose random phases are restricted to a subset of $[0, 2\pi]$ in the section titled “**Breaking of continuous time translation symmetry**” of the Supplementary Information. The supplementary content is nearly the same with that presented above. In the main text, we have also added several new sentences to discuss this,

on page 4, “...random phase distribution could arise from initial fluctuations. This distribution emerges due to the spontaneous breaking of continuous time translation symmetry, and not all self-oscillations in nonlinear systems exhibit this characteristic 2π random phase behavior (see Supplementary Information for more details)”.

Comment 3-3

2. The robustness of the oscillations against noise perturbations is attributed to the intrinsic properties of time crystals. However, this robustness is more accurately explained by the nonlinear negative feedback introduced through the experimental configuration. This feedback mechanism stabilizes oscillations and is a common feature in control systems and nonlinear oscillators.

Our response: We thank the reviewers for providing such an insightful perspective on understanding this phenomenon. Indeed, robustness is a common feature of feedback-induced nonlinear oscillators. The time crystals realized in our platform can be viewed as a special class of nonlinear systems and understood within the broader framework of nonlinear dynamics. Naturally, they inherit certain features associated with nonlinear systems, such as noise resilience. However, we would like to emphasize that not all nonlinear systems exhibit time-crystalline behavior. A defining feature of time crystals is not only their ability to support robust self-sustained oscillations, but also the spontaneous breaking of continuous time-translation symmetry—an essential hallmark that distinguishes them from conventional self-oscillatory systems.

Considering this, we have added several new sentences in two places of the main text, for example,

on page 5 in the main text, “*Having demonstrated the spontaneous breaking of continuous time-translation symmetry, we further verify the time crystalline nature of our system by examining its robustness against noise—a distinctive and desirable feature of time crystals that stems from the feedback-induced nonlinearity. We demonstrate this robustness by showing....*”.

In addition, we have added a new paragraph in the “**Discussion**” section,

on page 5 in the main text, “*Time crystals can manifest within nonlinear dynamical systems. However, not all nonlinear auto-oscillation behaviors qualify as CTCs. While CTCs share certain properties with general nonlinear systems, such as robustness against noise, it is the spontaneous symmetry breaking of continuous time translation symmetry that fundamentally distinguishes them...*”.

Comment 3-4

3. The transitions from limit cycles to quasi-periodic behavior, chaos, and back to limit cycles are presented as evidence of complex time crystal dynamics. However, these transitions are well-understood in the context of synchronization theory (see i.e. A. Pikovsky et al., Synchronization. A universal concept in nonlinear sciences (Cambridge University Press, Cambridge, 2010).):

- Limit cycles correspond to synchronized oscillations where the system locks onto a dominant frequency.
- Quasi-periodic behavior and chaos arise when synchronization breaks down due to mismatched frequencies or changes in feedback strength.

The reverse phase transition from chaos to limit cycles can also be explained through resynchronization phenomena, common in coupled oscillator systems (see Pikovsky again and A. Greilich et al., arXiv:2406.06243v1 as an example).

Our response: We thank the reviewer for pointing out the important connection to synchronization theory. This perspective is both insightful and encouraging, and we believe it will significantly aid future readers in understanding our work. Motivated by this suggestion, we reviewed relevant literature and were pleasantly surprised to find that our observed phenomena can indeed be well understood within this theoretical framework. Consequently, we have cited [A. Pikovsky et al., Synchronization. A universal concept in nonlinear sciences (Cambridge University Press, Cambridge, 2010); A. Greilich et al., Nat. Commun. 16, 2936 (2025)] in the revised manuscript to show this connection.

Considering this, we have added a new paragraph to the “**Discussion**” section to clarify the relationship between our observations and synchronization theory,

on page 5 in the main text, “...spontaneous symmetry breaking of continuous time translation symmetry that fundamentally distinguishes them. The rich spectrum of dynamical phases and the transitions observed in our system naturally arise from bifurcations within the nonlinear dynamics framework. Alternatively, these phenomena may be further interpreted with the synchronization theory, which offers a unifying perspective on complex collective behavior in coupled oscillator systems [A. Pikovsky et al., Synchronization. A universal concept in nonlinear sciences (Cambridge University Press, Cambridge, 2010); A. Greilich et al., Nat. Commun. 16, 2936 (2025)]. In our setup, the applied magnetic field gradient introduces a continuous distribution of Larmor frequencies ω , effectively treating each nuclear spin mode as an individual oscillator. These modes are globally coupled via the nonlinear feedback field (with strength α), while the gradient (characterized by g) imposes a frequency detuning across the ensemble—analogous to the mismatch of intrinsic frequencies in canonical synchronization models. The interplay between nonlinear coupling and detuning governs the emergence and breakdown of synchronization, giving rise to the observed dynamical phases. The limit cycle phase corresponds to a synchronized state with a dominant frequency, whereas quasi-periodic and chaotic dynamics emerge when synchronization is frustrated”.

Comment 3-5

4. As one can see, there is currently a tendency to name a many-body interacting system with nonlinear feedback mechanism, that leads to self-oscillations, a time crystal. The paper relies on feedback-induced oscillations which are related to self-oscillatory systems with parametric excitation. In such systems, the oscillations themselves control changes in system parameters through feedback, leading to complex dynamical behavior. An example of this is a swing, where the timing of the pumping force induces sustained oscillations. Therefore, I would make a careful introduction to your system and put it in perspective with previous studies on self-oscillatory systems.

Our response: We thank the reviewer for the thoughtful and constructive comment. Indeed, our system exhibits certain similarities to classical self-oscillatory systems, such as robustness against noise. On the other hand, not all self-oscillatory behavior constitutes a time crystal. A defining characteristic of our system is the spontaneous breaking of continuous time-translation symmetry, as evidenced by the random distribution of time phases from 0 to 2π across repeated

measurements (see the response to **Comment 3-3** for details). This hallmark sets our system apart from conventional self-oscillators and supports its classification as a continuous time crystal (CTC).

Considering this, we have added new sentences in several places in the main text and a new paragraph in “**Discussion**” section to introduce our system within the general framework of nonlinear dynamics and discuss the connection of the obtained results to time crystals. Most of the newly added content is presented in our response to **Comment 3-2**, **Comment 3-3** and **Comment 3-4**, with additional revisions as follows

on page 2 in the main text, “*The feedback-induced nonlinear terms, $\alpha P_z(\omega) \int \rho(\omega') P_x(\omega') d\omega'$ and $\alpha P_x(\omega) \int \rho(\omega') P_x(\omega') d\omega'$, can give rise to auto-oscillations that may exhibit collective synchronization, multi-frequency dynamics, or chaotic behavior. These phenomena can be understood within the general framework of dynamics of nonlinear oscillating systems [S.H. Strogatz, *Nonlinear Dynamics and Chaos*, Chapman and Hall/CRC, 2024; C. Sparrow, *The Lorenz Equations*, Springer, 1982; A. Pikovsky et al., *Synchronization. A universal concept in nonlinear sciences* (Cambridge University Press, Cambridge, 2010); A. Greilich et al., *Nat. Commun.* 16, 2936 (2025)]”.*

Comment 3-6

5. The current paper draws heavily on established principles from feedback-stabilized spin dynamics and nonlinear systems theory, as explored in the previous papers (Refs. [35,42,43]). The self-sustained oscillations, robustness due to feedback, and phase transitions between limit cycles, quasi-periodicity, and chaos are all phenomena well-documented in the spin maser literature and nonlinear dynamics. However, the framing of these behaviors as evidence of time crystals is the primary distinguishing factor of this paper. While this provides a novel perspective, it may overstate the originality of the findings, as the underlying physical mechanisms are consistent with those in previous studies.

Our response: We sincerely thank the reviewer for recognizing that our work “provides a novel perspective”. The studies mentioned by the reviewer (Refs. [35, 42, 43]) indeed form part of the background of our research and were accordingly cited in the our manuscript. However, our work goes beyond these studies. As already pointed out by the reviewer, one of the main contributions of our work lies in its perspective on time crystals. In addition to this, we would like to further emphasize that the experimental model and framework we investigate differ significantly from those previous works. Below, we provide an in-depth comparison with the articles:

- **Comparison to “Feedback assisted spin dynamics in an inhomogeneous bias magnetic field” (Ref. [35]):** Ref. [35] is purely theoretical work without performing validated experiments. While it addresses phenomena resembling those we observe, the spin models it employs differ from the conditions in our experimental setup. Specifically, the Ref. [35] study is based on a simple model, i.e., effective two-spin systems and triangular frequency distribution. In contrast, we propose and implement a new experimental scheme that provides the simplicity and control in platform via a gradient magnetic field. This leads to a distinct average spin frequency distribution, setting our experimental scenario apart from the models in Ref. [35]. Therefore, the findings of Ref. [35] are not directly applicable to our context.

- **Comparison to “Nuclear spin maser with an artificial feedback mechanism” (Ref. [42]) and “Floquet maser” (Ref. [43]):** We highlight two key differences between these previous studies and our work. Firstly, the underlying physical mechanisms of the studies differ. Refs. [42, 43] consider systems in which all spins evolve under identical Hamiltonians or dynamical equations. In contrast, our study considers dynamics governed by distinct equations for different spin frequency modes. Specifically, Ref. [43] focuses on the periodic driving and investigates the multiple Floquet energy levels formed by nuclear spins, while our work investigates spin dynamics in a two-level system. Secondly, the core aim of Ref. [42] was to develop a feedback technique that reduces the density requirement for self-sustained nuclear spin precession. It primarily contrasts the technique where a pick-up coil detects spin precession signals to generate a feedback magnetic field, which requires a large number of polarized nuclear spins. Although there is some technical continuity between Refs. [42, 43] and this work, the feedback technique is not the only core of this work. Instead, we discuss that our findings, though demonstrated on the present platform, may be extended to other systems and feedback mechanisms, such as those involving nuclear-spin liquids with pick-up coil feedback.

Considering this, we have added new sentences in several places in the main text:

on page 5, “ *It is worth noting that, while recent theoretical work (Ref. [Commun Phys 8, 41 (2025)]) explores similar phase transitions between limit cycles, quasi-periodicity, and chaos, it based on simple models that differ from our experimental conditions, leading to qualitatively different outcomes*”;

on page 5, “ *. . . the emergence of quasi-periodic and chaotic phenomena. While our demonstration focuses on a Rb-Xe hybrid gas system, we believe that integrating magnetic field gradients with feedback, as demonstrated in our work, is widely applicable across various platforms. This includes systems such as alkali-metal atomic gases, nuclear-spin liquids, and others. Furthermore, although we adopted the current feedback technique [Phys. Lett. A 304, 13 (2002); Sci. Adv. 7, eabe0719 (2021)], we believe our methodology is equally extensible to other feedback mechanisms*”.

Comment 3-7

6. While the paper employs chaos tests and the maximal Lyapunov exponent to characterize the system’s behavior, it lacks analysis of the spatial complexity of the attractors. Calculating the correlation dimension would provide deeper insights into the system’s dynamics. This additional metric would strengthen the claims regarding the nature of the observed dynamical phases.

Our response: We would like to thank the reviewer for this insightful comment. As suggested by the reviewer, analyzing the spatial complexity of the attractors indeed offers deeper insight into the system’s dynamics. Recently, pioneering work has been carried out in this aspect, for example, [A. Greilich et al., Nat. Phys. 20, 631 (2024)]. Following the methodology used in this work, we employ the TISEAN package [R. Hegger et al., Chaos 9, 413 (1999)] to compute the correlation dimension of our data. Below, we present our calculation results and analysis for the data corresponding to the three dynamical phases.

Figure 4 shows the calculation results of the correlation dimension of our data. The first row (i) in Fig. 4 presents the analysis of the limit cycle data (main text, Fig. 2d). Panel a displays

the log-log plot of the correlation sum C_2 versus the radius ϵ across increasing embedding dimensions m . Panel b depicts the slope of the curves from Fig. 4a, corresponding to the correlation dimension D_2 . The linear range of converged curves is observed around the ϵ range of 0.003 to 0.16, yielding $D_2 \approx 1.0 \pm 0.1$, indicated by the dark blue line. The integer value of D_2 reflects the periodic nature of the time series, consistent with the previously calculated $K \approx 0.0016$ and the absence of a positive Lyapunov exponent.

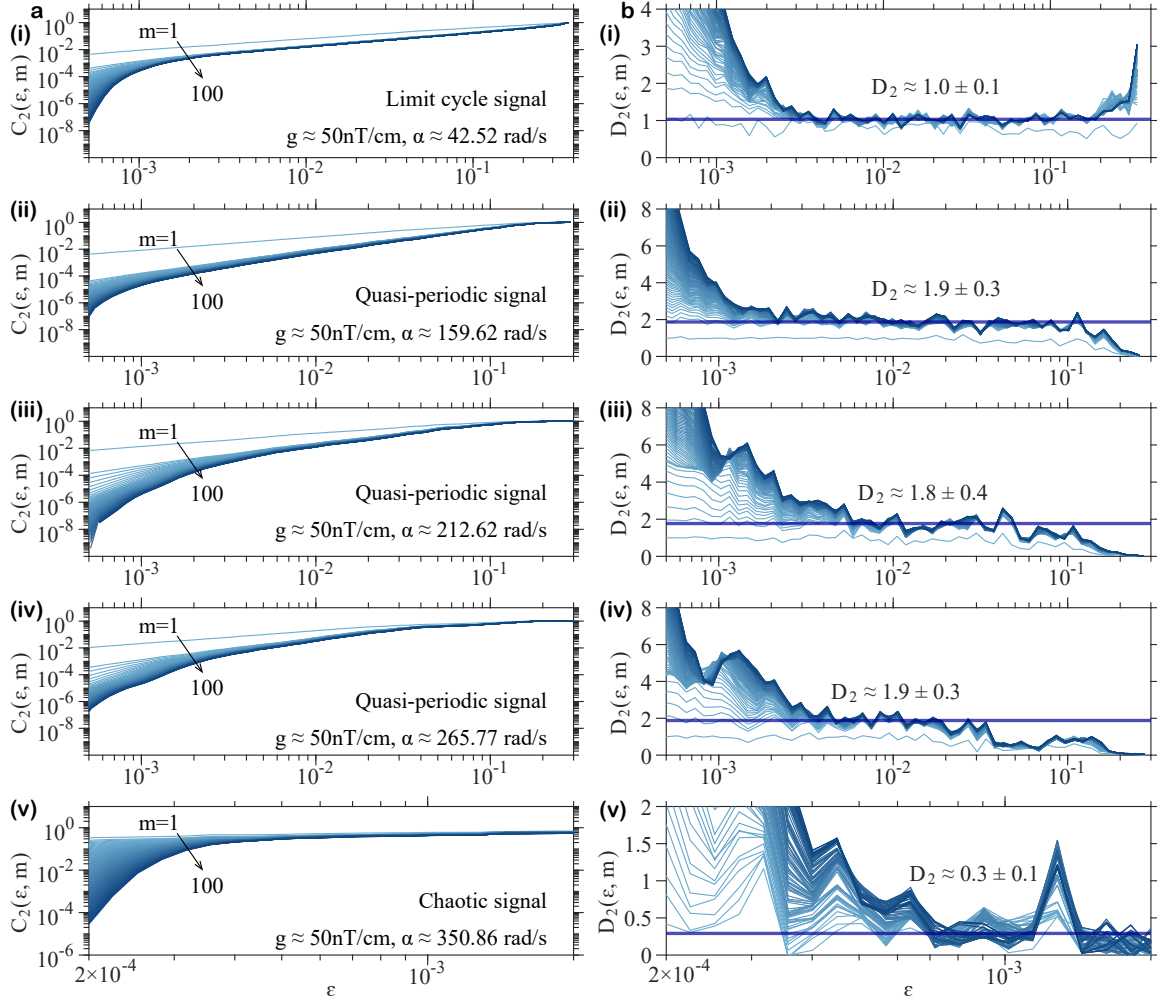


Figure 4: Calculation of correlation dimension. **a**, Log-log plot of the correlation sum C_2 versus increasing size of the threshold radius ϵ for an increasing number of embedding dimensions m . **b**, Plot of the slope in panel **a** as function of $\log(\epsilon)$, which converges to a constant value for a range of ϵ , shown by the dark blue line. The first row (i) presents the analysis of the limit cycle data. The rows (ii)-(iv) present the analysis of the quasi-periodic data. The last row (v) presents the analysis of the chaotic data.

The second row (ii) presents the analysis of the quasi-periodic data (main text, Fig. 2e). Using the same procedure, the computed correlation dimension is close to an integer value of 2 within the margin of error. Considering the diversity of the quasi-periodic phase, such as the varying number of spectral peaks under different experimental parameters, we analyze additional quasi-periodic data under various conditions, shown in rows (iii)-(iv). In all cases presented, the resulting correlation dimension are close to an integer value of 2.

The last row (v) shows the analysis of the chaotic data (main text, Fig. 2f). Applying the same

procedure, we obtain a non-integer correlation dimension of approximately $D_2 \approx 0.3 \pm 0.1$. This result further corroborates the chaotic nature, as indicated by the previously calculated $K \approx 0.9819$ and the Lyapunov exponent $\lambda \approx 0.35$.

As the reviewer pointed out, this additional metric strengthens the claims regarding the nature of the observed dynamical phases. Considering this, we have added a new subsection titled “**Correlation dimension**” in Supplementary Information to present the calculations and analysis of correlation dimension D_2 . The supplementary content is nearly the same with that presented above. In addition, to facilitate understanding among a broader readership, we have added a new paragraph in this subsection to introduce the fundamental principles of the D_2 calculation,

“The correlation dimension D_2 quantifies the fractal characteristics of a dynamical system’s attractor by describing how the number of point pairs within a given distance ϵ , denoted $C_2(\epsilon)$, scales with ϵ . For example, a simple limit cycle traces out a one-dimensional closed curve in phase space. At a given point on the limit cycle, the number of points within a spherical neighborhood of radius ϵ scales as $C_2(\epsilon) \sim \epsilon^1$ as ϵ increases, yielding a correlation dimension $D_2 \approx 1$. For quasi-periodic orbits evolving on an N -dimensional torus, the points uniformly fill an N -dimensional manifold, resulting in $D_2 \approx N$. In contrast, chaotic attractors typically exhibit non-integer correlation dimensions, reflecting their underlying fractal geometry and complex, self-similar structure in phase space. Given a reconstructed phase space $(\{\mathbf{s}_n\})$, the correlation dimension is estimated via the correlation sum:

$$C_2(m, \epsilon) = \frac{1}{N_{\text{pairs}}} \sum_{j=m}^N \sum_{k < j-w} \Theta(\epsilon - |\mathbf{s}_j - \mathbf{s}_k|), \quad (8)$$

where N_{pairs} is the number of pairs of points covered by the sums, $\Theta(\cdot)$ is the Heaviside step function that counts the number of pairs $(\mathbf{s}_j, \mathbf{s}_k)$ separated by less than ϵ , and w is the Theiler window that excludes temporally correlated points from the pair. On sufficiently small length scales and when the embedding dimension m exceeds the correlation dimension of the attractor,

$$C_2(m, \epsilon) \sim \epsilon^{D_2}, \quad (9)$$

such that a log-log plot of $C_2(m, \epsilon)$ versus ϵ yields a linear region whose slope provides an estimate of the correlation dimension D_2 . Since the correlation dimension is not known a priori, we check for convergence of the estimated values of D_2 in m . ”

Comment 3-8

Technical Questions and Clarifications:

7. Missing Methods Section: The manuscript lacks details on critical experimental parameters, which are essential for the reproducibility and understanding of the results:

- What are the pressures and temperatures of the Rb and Xe gases?
- What is the fractional ratio of the two constituents, and how does this influence Rb-Xe interactions?
- How does the oscillation frequency depend on parameters such as magnetic field strength, spin-exchange interactions, feedback strength, and field gradients?

While some of this information may be available in related works (e.g., arXiv:2302.05264), it

should be explicitly included in the current manuscript for completeness.

Our response: We thank the reviewer for his/her careful reading and help in pointing out the lack of description. We first address each issue raised in this comment point by point, and then provide additional technical details that may enhance the reader's understanding of our results:

- The vapor cell contains 50 Torr of ^{129}Xe (corresponding to a number density of $1.8 \times 10^{24} \text{ m}^{-3}$) and a droplet of ^{87}Rb , which provides saturated vapor. The cell is heated to 105°C , resulting in a saturated ^{87}Rb vapor number density of $8.3 \times 10^{18} \text{ m}^{-3}$.
- Given the respective number densities of ^{87}Rb and ^{129}Xe , the atomic number density ratio of Xe to Rb is approximately 2×10^5 . To deeply analyze the impact of the Xe–Rb fractional ratio on their interactions, we first briefly revisit the spin-exchange mechanism between Rb and Xe. The primary exchange processes involve binary collisions and the formation and destruction of van der Waals molecules via three-body collisions [see, for example, S. Applet et al., Phys. Rev. A 58, 1412 (1998); W. Happer et al., Phys. Rev. A 29, 3092 (1984)]. At low Xe gas pressures (well below one atmosphere), spin-exchange interactions are predominantly governed by three-body collisions. In the three-body collision scenario, an Rb atom and a Xe atom collide in the presence of a third body, in this case, a N_2 molecule or a Xe atom, forming a weakly bound van der Waals molecule. This molecule then evolves freely for a duration of τ_{vdw} before being disrupted by a subsequent collision. During the relatively long molecular lifetime, $\tau_{\text{vdw}} \approx 10^{-9} \text{ s}$, the weak coupling between the electron spin $\mathbf{S}$ of the alkali-metal atom, the rotational angular momentum $\mathbf{N}$ of the molecule, and the nuclear spin $\mathbf{K}$ of the noble-gas atom leads to the flipping down of $\mathbf{S}$ and partial flipping up of $\mathbf{K}$. Consequently, a fraction of the alkali metal spin polarization is transferred to the noble-gas nucleus. In binary collisions, as the lifetime of the binary collision, τ_{bc} , is very short ($\approx 10^{-12} \text{ s}$), only the nuclear spin $\mathbf{K}$ and the electron spin $\mathbf{S}$ couple, also resulting in the transfer of electron spin polarization to the nuclear spin. According to these mechanisms, both the binary collision rate and the three-body formation rate per Xe atom scale proportionally with the Rb number density $[\text{Rb}]$. Hence, the Xe spin-exchange optical pumping rate is directly proportional to $[\text{Rb}]$. The Xe number density $[\text{Xe}]$ primarily affects the spin-exchange interaction by participating as the third body in the van der Waals molecule formation process. On one hand, a higher $[\text{Xe}]$ enhances the formation rate of van der Waals molecules; on the other hand, it reduces their lifetime τ_{vdw} . Typically, the negative impact of latter outweighs the benefits of the former. From the perspective of spin-exchange rate, a higher Rb density and a lower Xe density seems favorable. However, increasing the Rb density also accelerates Xe spin relaxation due to collisions, and an insufficient Xe density undermines signal readout sensitivity. Therefore, the Rb number density should not be too high, nor should the Xe number density be too low. Determining the optimal Xe-Rb fractional ratio theoretically is complex. In this experiment, based on experience, we have chosen a ^{129}Xe gas pressure of 50 Torr, with the ^{87}Rb saturated vapor temperature set to 105°C .
- This is a very important question. The fundamental oscillation frequency mainly depends on the mean magnetic field which sets the central Larmor frequency for spins. This fundamental frequency can slightly shift, and additional frequency components may emerge due to nonlinear interactions within the spin ensemble. The shift in the fundamental frequency and the frequency of these additional components are influenced by system parameters. Previous research, such as [Commun Phys 8, 41 (2025)], has explored a simplified two-spin model using dimensional reduction and numerical simulations to map

quasi-periodic frequency intervals against parameters like feedback strength α . However, in our system, where a linear gradient magnetic field is applied, the system corresponds to an infinite-frequency mode spin coupling, introducing significant complexity. The findings from the two-spin model cannot be extrapolated to this case. While our current experiments (Fig. 3 of the main text) preliminarily explore this by displaying Fourier spectra under varying feedback and gradient strengths, a comprehensive theoretical framework for frequency dependence in this infinite-mode regime remains an unresolved issue.

Inspired by the reviewer’s comment, we have incorporated additional technical details into the “**Experimental Details**” section of the Supplementary Information and added a new section titled “**Analysis of parameter dependence**” to discuss parameter dependencies and explore alternative feedback configurations. The revised section “**Experimental Details**” is now organized into four subsections: **A. Spin system; B. Spin exchange optical pumping; C. Optical probe; D. Feedback**. The content of the subsections “**Optical probe**”, “**Feedback**”, and the relevant portion of “**Analysis of parameter dependence**” will be addressed in detail in response to **Comment 3-9**, **Comment 3-10**, **Comment 3-11** and **Comment 4-2**, respectively. The remaining supplementary content is nearly the same with that presented above.

Comment 3-9

8. Optical Detection Setup: The combination of the $\lambda/4$ plate, PEM, and linear polarizer (LP) in the detection scheme requires further clarification:

- How is the PEM aligned relative to the fast axis of the $\lambda/4$ plate?
- What retardation is used for the PEM—is it also set to $\lambda/4$?
- Does this setup mean you are primarily measuring the Faraday ellipticity signal rather than Faraday rotation? Why was this configuration chosen?
- I think a balanced photodiode setup would improve polarimetric sensitivity. Why was it not implemented?

Our response: We thank the reviewer for his/her careful reading and pointing out the lack of description. We begin by briefly summarizing our responses to the questions raised in the comments, and then provide a systematic introduction to our optical detection setup.

- The PEM is aligned at 45 degrees relative to the fast axis of the $\lambda/4$ plate.
- The PEM is not set to a fixed retardation like $\lambda/4$; instead, it introduces a time-dependent phase retardation described by:

$$A(t) = A_0 \sin \Omega_m t, \quad (10)$$

where A_0 is the modulation amplitude, typically near a few radians (e.g., 2.405 rad to optimize 1st-order Bessel function), Ω_m is the modulation frequency (e.g., 50 kHz).

- Yes, we ultimately read out the ellipticity modulation signal of light, but in this case, this signal is directly related to the Faraday rotation angle. This configuration is intended to achieve a higher signal response following PEM modulation. When the light entering the PEM is linearly polarized, only even-order modulation frequency terms (e.g., $2\Omega_m$) appear in the detection signal, and the signal-to-noise ratio is poor. The $\lambda/4$ wave plate convert Faraday rotation into an ellipticity modulation, causing the detection signal to

contain odd-order modulation frequency terms (e.g., Ω_m), which significantly improves the sensitivity. Detailed derivations are provided later in this comment.

- Indeed, the balanced photodiode setup is a commonly used configuration and offers high detection sensitivity. In principle, it could also be adopted in our setup. However, there are some low-frequency noises (covering DC-100 Hz) in the experiment, such as 1/f noise from photodetectors, airflow, mechanical vibrations of optical components, and fluctuations in laser power and wavelength. In order to suppress the influence of low-frequency noise of the probe beam, we employ a PEM to modulate the polarization of probe beam at a high frequency (50 kHz).

To provide a clear description of the detection scheme and specific detection principles, we have added a new subsection of “**Optical Detection**” in Supplementary Information. The supplementary content is as follows:

“The spin polarization of ^{129}Xe is measured via the electron spin polarization $\mathbf{P}^e$ of ^{87}Rb atoms through spin exchange interaction. The transverse electron spin polarization of ^{87}Rb along x is measured through the optical rotation of the probe beam tuned to the D2 transition of ^{87}Rb [Z. Wu et al., Appl. Opt. 25, 4483 (1986); W. Opechowski, Rev. Mod. Phys. 25, 264 (1953)]:

$$\theta = \frac{1}{4} l r_e c f_{D2} [Rb] D(\nu_{pr}) P_x^e, \quad (11)$$

where $l \approx 0.8 \text{ cm}$ is the optical path length, $r_e = 2.8 \times 10^{-13} \text{ cm}$ is the classical radius of the electron, c is the speed of light, $f_{D2} \approx 2/3$ is the oscillator strength, $[Rb]$ is the number density of ^{87}Rb atoms, $D(\nu_{pr}) = (\nu_{pr} - \nu_{D2}) / [(\nu_{pr} - \nu_{D2})^2 + (\Delta\nu_{D2}/2)^2]$, ν_{pr} is the frequency of the probe laser light, $\Delta\nu_{D2}$ is the full-width at half-maximum (FWHM) of the optical D2 transition of frequency ν_{D2} . In this experiment, the probe beam power is approximately 2 mW.

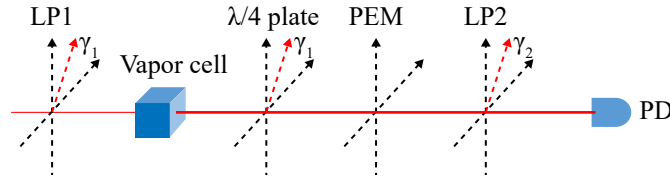


Figure 5: Optical path diagram of the photoelastic modulation scheme. The optical axis of the PEM is aligned with the horizontal axis (x-axis). The fast axis of the quarter-wave ($\lambda/4$) plate is aligned parallel to the transmission axis of the first polarizer (LP1). γ_1 and γ_2 denote the angles between the polarization axes of LP1 and LP2 and the x-axis, respectively.

The Faraday rotation angle is typically very small and requires a precise detection setup. Considering the low-frequency noise (covering DC-100 Hz) in experiment, such as 1/f noise from photodetectors, airflow, mechanical vibrations of optical components, and fluctuations in laser power and wavelength, a photoelastic modulator (PEM) is used to shift the low-frequency Faraday rotation angle signal into a higher frequency range (specifically, 50 kHz in this case). The PEM is a phase modulator based on the photoelastic effect in a crystal, where the phase difference between its fast and slow axes varies periodically under cyclic mechanical pressure: $A(t) = A_0 \sin \Omega_m t$, where A_0 is the modulation amplitude, typically near a few radians (e.g., 2.405 rad to optimize 1st-order Bessel function), Ω_m is the modulation frequency (e.g., 50 kHz). As the probe light passes through the crystal, its phase undergoes periodic modulation, shifting of the originally low-frequency optical signal into a higher frequency range. The experimental

optical path is shown in Fig. 5. A linearly polarized probe beam passes through a first polarizer, an atomic vapor cell, a $\lambda/4$ wave plate, a PEM and a second polarizer in sequence, and finally enters the photodiode. The optical axis of the PEM is aligned with the x-axis. The fast axis of the quarter-wave plate is aligned parallel to the transmission axis of the first polarizer (LP1). Assuming that the angles between the polarization axes of the first and second polarizers and the x-axis are denoted by γ_1 and γ_2 , respectively. In the following, we calculate the probe light intensity after it traverses the optical path depicted in Fig. 5 using Stokes vectors and the Mueller matrix.

The Stokes vector of the incident light passing through LP1 is expressed as:

$$\mathbf{S}_1 = I_0 \begin{bmatrix} 1 & \cos 2\gamma_1 & \sin 2\gamma_1 & 0 \end{bmatrix}^T. \quad (12)$$

After passing through the cell, it transforms to:

$$\mathbf{S}_2 = M_{cell}(\theta) \cdot \mathbf{S}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\theta & \sin 2\theta & 0 \\ 0 & -\sin 2\theta & \cos 2\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot I_0 \begin{bmatrix} 1 \\ \cos 2\gamma_1 \\ \sin 2\gamma_1 \\ 0 \end{bmatrix} = I_0 \begin{bmatrix} 1 \\ \cos(2\gamma_1 - 2\theta) \\ \sin(2\gamma_1 - 2\theta) \\ 0 \end{bmatrix}. \quad (13)$$

Subsequently, passing through a quarter-wave plate modifies it to:

$$\begin{aligned} \mathbf{S}_3 = M_{\lambda/4}(\gamma_1) \cdot \mathbf{S}_2 &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos^2 2\gamma_1 & \sin 2\gamma_1 \cos 2\gamma_1 & -\sin 2\gamma_1 \\ 0 & \sin 2\gamma_1 \cos 2\gamma_1 & \sin^2 2\gamma_1 & \cos 2\gamma_1 \\ 0 & \sin 2\gamma_1 & -\cos 2\gamma_1 & 0 \end{bmatrix} \cdot I_0 \begin{bmatrix} 1 \\ \cos(2\gamma_1 - 2\theta) \\ \sin(2\gamma_1 - 2\theta) \\ 0 \end{bmatrix} \\ &= I_0 \begin{bmatrix} 1 \\ \cos 2\gamma_1 \cos 2\theta \\ \sin 2\gamma_1 \cos 2\theta \\ \sin 2\theta \end{bmatrix}. \end{aligned} \quad (14)$$

Then the PEM modulate it as:

$$\begin{aligned} \mathbf{S}_4 = M_{PEM}(A(t), 0^\circ) \cdot \mathbf{S}_3 &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos A(t) & \sin A(t) \\ 0 & 0 & -\sin A(t) & \cos A(t) \end{bmatrix} \cdot I_0 \begin{bmatrix} 1 \\ \cos 2\gamma_1 \cos 2\theta \\ \sin 2\gamma_1 \cos 2\theta \\ \sin 2\theta \end{bmatrix} \\ &= I_0 \begin{bmatrix} 1 \\ \cos 2\gamma_1 \cos 2\theta \\ \cos A(t) \sin 2\gamma_1 \cos 2\theta + \sin A(t) \sin 2\theta \\ -\sin A(t) \sin 2\gamma_1 \cos 2\theta + \cos A(t) \sin 2\theta \end{bmatrix}. \end{aligned} \quad (15)$$

The Mueller matrix for LP2 is:

$$M_{LP}(\gamma_2) = \frac{1}{2} \begin{bmatrix} 1 & \cos 2\gamma_2 & \sin 2\gamma_2 & 0 \\ \cos 2\gamma_2 & \cos^2 2\gamma_2 & \sin 2\gamma_2 \cos 2\gamma_2 & 0 \\ \sin 2\gamma_2 & \sin 2\gamma_2 \cos 2\gamma_2 & \sin^2 2\gamma_2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}. \quad (16)$$

After passing through LP2, the output light intensity is the first component of the Stokes vector after the action of the LP2 Mueller matrix:

$$I_{out} = \frac{I_0}{2} [1 + \cos 2\gamma_2 \cos 2\gamma_1 \cos 2\theta + \sin 2\gamma_2 (\cos A(t) \sin 2\gamma_1 \cos 2\theta + \sin A(t) \sin 2\theta)] \quad (17)$$

Since the Faraday rotation angle θ is very small, we apply the small-angle approximations $\cos \theta \approx 1$ and $\sin \theta \approx \theta$, thus :

$$I_{out} \approx \frac{I_0}{2} [1 + \cos 2\gamma_1 \cos 2\gamma_2 + \sin 2\gamma_1 \sin 2\gamma_2 \cos A(t) + 2\theta \sin 2\gamma_2 \sin A(t)] \quad (18)$$

The terms $\cos A(t) = \cos(A_0 \sin \Omega_m t)$ and $\sin A(t) = \sin(A_0 \sin \Omega_m t)$ can be expanded into Bessel function series. Given that A_0 is small, expanding up to second order yields the approximations: $\cos A(t) \approx 1 - \frac{A_0^2}{4} + \frac{A_0^2}{4} \cos 2\Omega_m t + O(A_0^4)$, $\sin A(t) \approx A_0 \sin \Omega_m t + O(A_0^3)$. Thus, the output light intensity is approximately:

$$I_{out} \approx \frac{I_0}{2} [1 + \cos(2\gamma_1 - 2\gamma_2)] + \theta A_0 I_0 \sin 2\gamma_2 \sin \Omega_m t - \frac{A_0^2}{8} I_0 \sin 2\gamma_1 \sin 2\gamma_2 \cos 2\Omega_m t. \quad (19)$$

From above equations, we can determine that the desired signal is directly proportional to the first-order modulation frequency component.

- To maximize the first harmonic, $\sin 2\gamma_2 = \pm 1$ should be satisfied;
- To minimize the DC signal, $\cos 2(\gamma_1 - \gamma_2) = -1$ should be satisfied.

To maximize the desired signal, we set $\gamma_2 = \pm\pi/4$, $\gamma_1 = \mp\pi/4$.

Comment 3-10

9. Exploration of Alternative Feedback Configurations: Have the authors explored other types of feedback configurations?

- For example, using the pump power or polarization instead of the magnetic field feedback.
- Applying feedback along different magnetic field directions—would this affect the dynamical phases observed?

Our response: We sincerely appreciate the reviewer's constructive questions, which inspire us to further explore alternative feedback configurations in our future research. Below, we address each issue raised in this comment point by point:

- The magnetic field feedback enters the Bloch equations through a cross product with the polarization vector, introducing nonlinearity due to the product terms of polarization components. When the magnetic field feedback is removed and replaced by feedback proportional to the average polarization via the pump power or polarization, no nonlinear terms arise. However, in the presence of feedback magnetic field, modulating the pumping or polarization could potentially lead to additional phenomena—such as synchronization induced by external driving—similar to those observed in Ref. [A. Greulich et al., Nat. Commun. 16, 2936 (2025)]. A detailed exploration of such effects lies beyond the scope of the present work and is probably left for future investigation.
- In the main text, the feedback field is applied along the y -axis, given by $B_y = -\alpha \bar{P}_x / \gamma$. According to the reviewer's comment, we perform new simulations with the feedback field applied along the x -axis, i.e., $B_x = -\alpha \bar{P}_x / \gamma$. These simulations reveal that the system evolves toward a steady state, characterized either by vanishing transverse polarization or by a non-zero constant transverse polarization. Two representative examples are shown

in Fig. 6: panel a illustrates a case where the system relaxes to a state with $\bar{P}_x = 0$, while panel b shows the system stabilizing at a finite, constant $\bar{P}_x$.

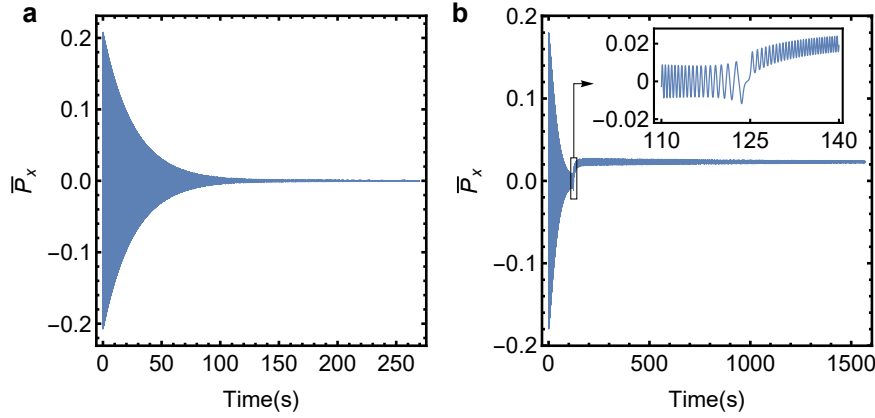


Figure 6: Representative dynamics with the feedback field applied along the x -axis, i.e., $B_x = -\alpha\bar{P}_x/\gamma$. The parameters are chosen as follows: $\alpha = 76.92$ rad/s, $g = 3.89$ nT/cm for **a**; and $\alpha = -769.23$ rad/s, $g = 3.89$ nT/cm for **b**.

Considering this, we have added a new discussion about alternative feedback configurations in the newly added section “**Analysis of parameter dependence**” in the Supplementary Information. The supplementary content is nearly the same with that presented above.

Comment 3-11

10. Power Dependence of the Pump and Probe Beams:

- What are the powers (or power densities) of the pump and probe beams?
- Is there any dependence of the oscillation behavior on the beam power? This could influence the spin polarization and feedback strength, affecting the system dynamics.

Our response: We really appreciate the reviewer for pointing out these insightful comments. As noted by the reviewer, the oscillation behavior depends on the beam power. In order to clearly demonstrate this dependence, we have incorporated additional technical details. For example, in the revised section “**Experimental details**” of the Supplementary Information, we specify that in our experiment, the pump and probe laser powers are 11.5 mW and 2 mW, respectively. Furthermore, in the newly added section “**Analysis of parameter dependence**”, we have included a brief discussion on the dependence of the oscillation behavior on beam power. For example, altering the pump beam power changes the absorption rate of Rb atoms for photons, thereby affecting the spin polarization of ^{87}Rb . This influences both the polarimetric sensitivity that impacts the feedback strength α , and the spin exchange between Rb and Xe. Both factors influence the oscillation behavior of ^{129}Xe . Similarly, adjusting the probe beam power also affects polarimetric sensitivity, subsequently impacting the feedback strength α , and ultimately influencing the oscillation behavior of ^{129}Xe .

Comment 3-12

The paper contributes to the ongoing exploration of nonlinear spin dynamics and precision measurement technologies, and its framing within the time crystal paradigm may stimulate further interdisciplinary research. However, the claims regarding the discovery of new phases of matter should be tempered with a deeper acknowledgment of the classical origins of the observed behaviors. I recommend major revisions to address the concerns outlined above. Specifically, the interpretation of the results should be reconsidered in light of established knowledge from nonlinear dynamics and feedback-stabilized spin systems. Additionally, more technical details and systematic investigations of parameter variations would strengthen the paper. At this point it is hard for me to accept the paper for a publication in Nature Communications. It is too similar to the previous research, partially by the same authors, see Refs. [35, 43], and requires more than just a rebranding of the observed oscillations. I will reassess the work upon the authors' response to these points and the feedback from other reviewers.

Our response: We sincerely thank the reviewer for recognizing the significance of our work in advancing the understanding of nonlinear spin dynamics and its potential to inspire interdisciplinary research through the lens of time crystals. In response to the reviewer's thoughtful comments, we have made major revisions to the manuscript, particularly regarding the identification and interpretation of new dynamical phases associated with continuous time crystals. Below, we summarize the main changes in the manuscript:

- We describe our system within the framework of feedback-induced auto-oscillators, which exhibit common features of nonlinear dynamical systems, such as robustness and random oscillation phases. We further clarify the distinction between our CTCs and normal auto-oscillators by examining whether they truly exhibit spontaneous breaking of continuous time-translation symmetry—that is, whether the randomness in phase spans the entire range from 0 to 2π , rather than being confined to a narrow interval.
- We incorporate synchronization theory to interpret the observed dynamical transitions and cited relevant foundational works to frame our findings within a broader theoretical context.
- We compare our work with the previous research mentioned by the reviewer to highlight its novelty.
- We supplement a more systematic analysis and included additional experimental details. For example, we have added new subsections in the Supplementary Information to provide experimental details about our spin system, spin-exchange optical pumping, the optical detection setup, and the feedback configuration. Additionally, we have included a brief discussion on the influence of key experimental parameters, such as gas composition and beam power, on the spin interactions and oscillation behavior.
- We explored the possibility of alternative feedback configurations through additional simulations and analysis, further strengthening the completeness of our study.

We would like to thank again the reviewer for his/her careful reading and insightful suggestions, which have greatly improved the manuscript. We hope that the reviewer now find the paper suitable for Nature Communications.

Reply to Reviewer #4

Comment 4-1

The manuscript by Huang and collaborators reports on the experimental observation of continuous time crystals and continuous time quasi-crystals in a gas of spin particles (Xe, Rb). Using a feedback loop, namely acting on the system with a field whose strength depends on the system itself, the authors create a nonlinearity that can lead to a rich phase diagram. As one could expect from nonlinear ODEs of three variables (Eq. 1), tuning the parameters can lead to (i) limit cycles, (ii) oscillations at multiple incommensurate modes, and (iii) chaos. The authors argue for continuous time translational symmetry breaking, which elevates (i) to the status of continuous time crystal (CTC) and (ii) to that of continuous time quasi-crystal (CTQC). Robustness to noise is checked, and potential applications for metrology are argued for.

I find the paper interesting, relatively well written, and with the potential to have an impact in the field of time crystals. Before making a recommendation towards publication in Nature Communications, however, I would ask the authors to address a key point on what elevates a limit cycle to a CTC, together with a couple of other minor comments.

Our response: We would like to thank the reviewer for his/her careful reading and potential recommendation of our work. The reviewer's constructive suggestions help us to improve the quality of our manuscript. In particular, as raised by the reviewer concerning the random phase, we have added a new subsection in the Supplementary Information to quantitatively explain how quickly the feedback can be activated relative to the period of the CTC, thereby excluding it as a potential technical reason for the randomness in the CTC phase. Additionally, we have included new simulated time trace graphs to more clearly illustrate how the phase of oscillations changes from run to run as a result of quantum fluctuations. On the basis of these comments, we have carefully revised our main text and Supplementary Information. Below, we provide the detailed reply to the comments and the corresponding major changes in the revised paper.

Comment 4-2

Following [Science 377.6606, 670-673 (2022)], the authors argue that the key ingredient that elevates a limit cycle to the CTC state is that, upon repeating the experiment multiple times under identical conditions, the phase of the oscillations is different for every run, with a uniform distribution between 0 and 2π . In the experiment, a key action is the activation of the feedback. If the activation time is random, so will be the phase of the oscillations, but such randomness of the phase would follow from that of the feedback activation and thus not be indicative of CTC. The question is thus: can the feedback be activated quickly and precisely (with respect to the period of the CTC), so that it can be excluded as an obvious reason for the randomness of the CTC phase? A couple of numbers to quantify "quickly" and "precisely" in the previous sentence would be of help.

Our response: We would like to thank the reviewer for pointing out the lack of description. As pointed out by the reviewer, the temporal precision of feedback activation is crucial in determining the relative time phase. In fact, this factor was a central consideration in our experimental design. To achieve high-precision control over feedback activation timing, we carefully designed the feedback switch control circuit and synchronized the data acquisition timing, while also experimentally calibrating all potential sources of timing error. Below, we provide additional experimental details to quantify how these techniques influence the phase.

We begin by detailing the feedback design. In the experiment, the probe signal, which contains information about spin polarization, is captured by a photodetector. After demodulation with a lock-in amplifier, the signal is converted into a voltage signal V . This voltage is then fed through a series resistor R to y-direction coils, thereby generating a feedback magnetic field B_f . To control the activation of the feedback, a relay is incorporated into the circuit. The relay switch is controlled by a TTL digital output module (NI 9401), which outputs a digital voltage of either 0 or 5 V. When the TTL signal is high, the relay connects, activating the feedback; when the signal is low, the relay disconnects, deactivating the feedback. For consistent determination of the relative time phase across different experimental runs, a common time origin is necessary. Both the TTL signal and the probe signal are synchronously recorded using a 4-channel, 24-bit simultaneous analog input module (NI 9239). The rising edge of the TTL signal is designated as the temporal zero point. Notably, the NI 9239 is a simultaneous sampling ADC module, ensuring no inter-channel delay.

The key parameters of the feedback system that influence the relative time phase. Critical factors affecting timing accuracy is the rising edge timing of the TTL signal, denoted as τ_{up} . Based on our 20 experimental calibrations, we determined that the rising edge τ_{up} of the TTL signal does not exceed 0.04 ms. This level of timing uncertainty has a negligible impact on the time phase distribution. For example, in comparison to the period of the CTC (approximately 57.1 ms at a frequency of 17.5 Hz), any potential phase shift caused by these uncertainties is at most $0.04/57.1 \times 2\pi \approx 0.0014\pi$. This is far too small to account for the observed 2π random phase distribution.

Considering this, we have added several new sentences in the main text to highlight the precision in the timing of feedback activation:

on page 4 in the main text, *“This confirms the spontaneous breaking of continuous time-translation symmetry in the limit cycle phase. Accurate feedback activation is essential for determining the relative time phase. To achieve this, we design a high-precision feedback switch control circuit along with synchronized data acquisition (see Supplementary Information for details)”*.

In addition, we have added a detailed description about the feedback activation in the subsection **“Feedback”** in the Supplementary Information. The supplementary content is nearly the same with that presented above.

Comment 4-3

Related to the previous point. In the end, of course in the experiments there will be many sources of noise, and these can be amplified to ultimately determine the phase of the CTC. Yet, I like the analysis in Supplemental Section I, where the authors argue, even just as a matter of principle, that the randomness of the phases can result from the quantum fluctuations alone. The plot in Fig. 1 is good because it connects to the figures in the main, as well as to the figures in [Science 377.6606, 670-673 (2022)], where the concept of CTC was established. Yet, it would be of help to see some time traces. Presumably the time traces will show that oscillations are developed dynamically, and that the phase of these oscillations changes from run to run.

Our response: We thank the reviewer for his/her insightful suggestion. As suggested by the reviewer, we have included several time traces in the Supplemental Figure within the Supplementary Information (see below). These time traces correspond to three distinct dynamical

phases, illustrating how oscillations develop dynamically and how the phase of these oscillations varies from run to run. Below, we provide further details on the time traces across the three different dynamical phases.

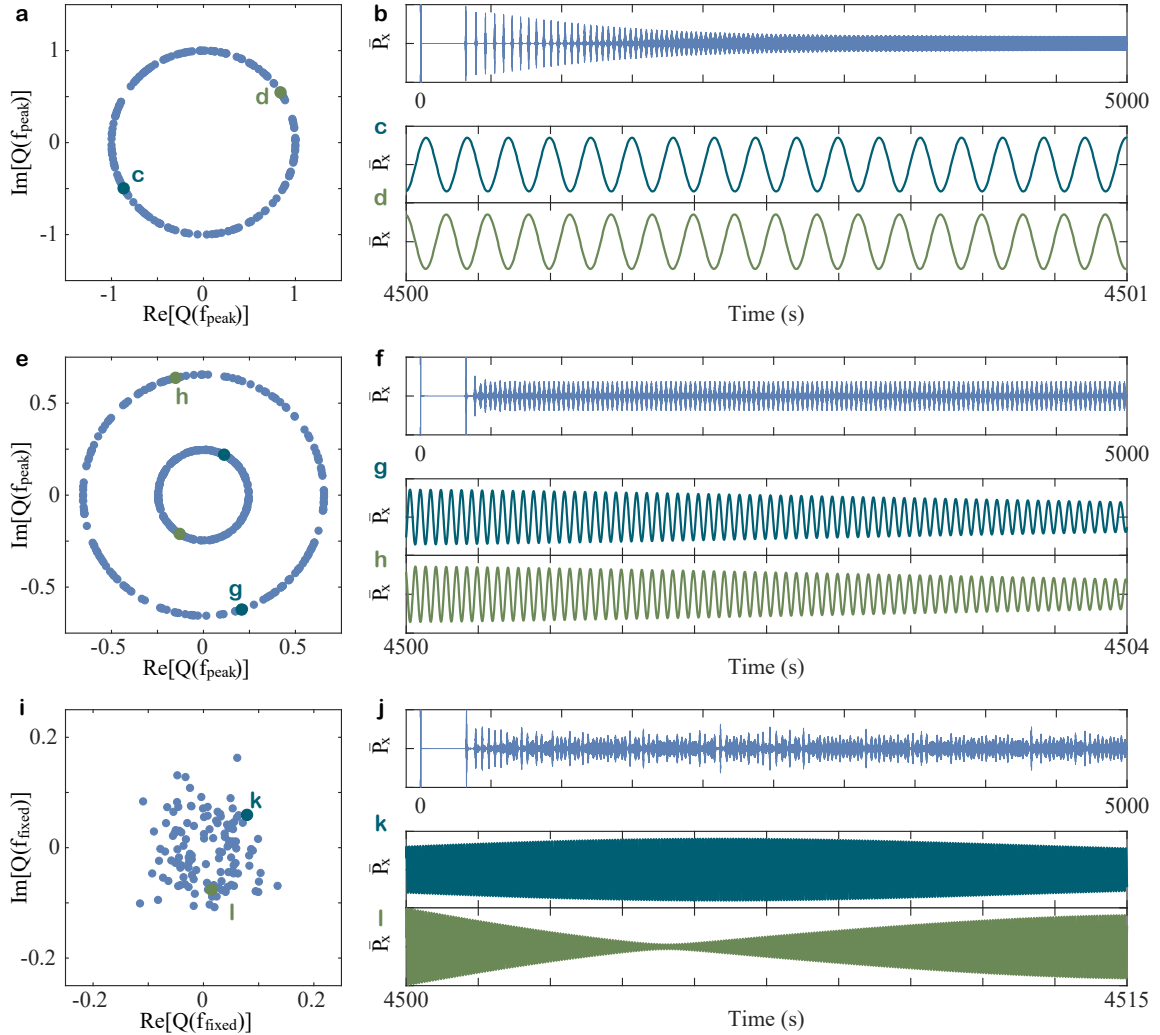


Figure 7: Phase distributions and corresponding time traces for limit cycle (a–d), quasi-periodic (e–h), and chaotic oscillations (i–l).

(1) **Limit cycle.** Figure 7a presents the simulated time phases of the limit cycle when only the initial quantum fluctuations are considered, displaying a random distribution covering $[0, 2\pi]$. Each point represents an individual simulation. Figure 7b illustrates the typical dynamical evolution of a single simulation, where oscillations spontaneously develop and then stabilize. We perform a Fourier transform of the spin polarization $\bar{P}_x$ during the steady-state regime and extract the complex Fourier amplitude at the peak frequency, $Q(f_{\text{peak}})$. To clearly illustrate how the phase of these oscillations varies from run to run, we select two specific $Q(f_{\text{peak}})$ points from Fig. 7a (labeled “c” and “d”) which have a time phase difference of almost π , and show their corresponding time traces $\bar{P}_x(t)$ in Figs. 7c and d, respectively.

(2) **Quasi-periodic oscillation.** Similarly, Figures 7e–h present the case of quasi-periodic oscillations. Unlike the limit cycle, the quasi-periodic oscillation features multiple spectral peaks. Several prominent frequency components are shown in Fig. 7e, and as a result, each individual simulation corresponds to multiple points in the plot. The dynamical evolution of

the points in Fig. 7e is exemplified in Fig. 7f, where the signal exhibits wave packets with stable envelopes. We select two points labeled “g” and “h”, which exhibit a phase difference of almost π at the dominant frequency (the one with the highest peak), for detailed comparison. Since the phase difference between “g” and “h” is not apparent on the time scale of the full wave packet, we zoom in to examine the internal structure of a single packet. As shown in Fig. 7g and h, while the two signals exhibit similar amplitude modulation in envelopes, their underlying carrier oscillations are nearly in opposite phase at any given moment.

(3) **Chaotic phase.** For comparison, the distribution of simulated time phases in the chaotic regime is depicted in Fig. 7i, where all points scatter across a circular surface. This scattering arises from the lack of stable oscillations in chaos, resulting in significant variability in the Fourier amplitude at a fixed frequency across repeated simulations. The dynamical evolution of the points in Fig. 7i is exemplified in Fig. 7j, where the signal exhibits wave packets with varying envelopes. To illustrate the phase variability, we select two points labeled “k” and “l” which show a large phase difference at a fixed frequency, and display their respective time traces. At a specific moment, when the oscillation in Fig. 7k reaches a peak of a wave packet, the corresponding signal in Fig. 7l is near a zero crossing of a wave packet.

Considering this, we have added new simulated time trace graphs and several new paragraphs in the section titled “**Breaking of Continuous Time Translation Symmetry**” of the Supplementary Information. The supplementary content is nearly the same with that presented above.

Comment 4-4

Random phase with respect to the phase of the feedback switch, or to the drive phase. In particular, how fast is that compared to the drive frequency? The simulation in Supplementary Material is particularly interesting. I would like to see various time traces, the transient buildup of oscillations, the difference in phases from run to run.

Our response: Indeed, this is very important question. The random phase distribution is a core criterion of CTC. Therefore, it is crucial to distinguish whether it originates from technical issues or intrinsic quantum fluctuations. We address this issue from both perspectives:

Regarding technical noise, to prevent the random phase from arising due to technical timing errors, we precisely control the feedback activation to ensure it operates quickly and precisely relative to the CTC period. More detailed information can be found in [Comment 4-2](#).

Regarding quantum noise, to provide a clearer understanding of the phase differences caused by quantum fluctuations, we have included additional time traces from various runs in Fig. S1 of the revised Supplementary Information. More detailed information can be found in [Comment 4-3](#).

Comment 4-5

Further minor comments

1. The authors mention a couple of experimental works on discrete time quasi-crystals. It could be worth mentioning the theory works, too. In particular, the idea that a magnetic field gradient can induce a second frequency, that is gradient-dependent and incommensurate, leading to quasi-crystallinity, was introduced for the Floquet case in [PRL 123.15, 150601(2019)].

Our response: We really appreciate the reviewer for his/her help in pointing out our omission of this important theoretical work [A. Pizzi, J. Knolle, and A. Nunnenkamp, PRL 123, 150601(2019)]. This study provides a valuable perspective on the concept of time quasicrystal. Considering this, we have cited this work in the updated reference list and referenced it as [40]. In addition, we have also added a new sentence to introduce this important work,

on page 4 in the main text, “...characteristic of recently proposed time quasi-crystal. As an example, Ref. [40] revealed the emergence of time quasicrystal in the presence of a “staircaselike” on-site potential”.

Comment 4-6

2. This is more of a stylistic matter, but I find the text bombastic and empty at times. For instance, I find the sentence in the abstract pretty empty: “However, their complex properties still require further exploration, and many phenomena remain to be revealed”. Adjectives such as “exciting”, “new”, “novel”, and “significant” could be toned down.

Our response: We thank the reviewer for pointing out the stylistic issues in our manuscript. According to the reviewer’s suggestions, we have carefully modified our manuscript and removed these overly subjective or imprecise descriptions.

Additionally, we would like to thank again the reviewer for his/her careful reading and approval of our work. On the basis of these insightful comments, we have made changes to improve the quality of our paper. We believe that we have addressed well the reviewer’s comments, and hope that the reviewer now finds the paper suitable for Nature Communications.

Reply to Reviewer #1

Comment 1-1

The authors have addressed all my concerns and comments thus I am happy to recommend the work for publication in Nature Communications.

Our response: We sincerely thank the reviewer for the recommendation.

Reply to Reviewer #2

Comment 2-1

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Our response: We really appreciate the reviewer's co-review.

Reply to Reviewer #3

Comment 3-1

I have carefully read the authors' replies to my report and those of the other referees. I find the responses satisfactory and appreciate the substantial revisions and clarifications added to the manuscript, especially regarding the dynamical regimes, phase diagrams, and the connections to synchronization theory and nonlinear dynamics.

My main remaining comment is that the authors should be more careful in assigning the continuous time crystal (CTC) label based solely on the observation that the phase of oscillations covers the entire circle in repeated measurements. As demonstrated in [Kongkhambut, P. et al., Rep. Prog. Phys. 87, 080502 (2024)], when a system is subject to additional periodic driving, the phase distribution can become non-uniform or concentrated on particular sectors of the circle, rather than being randomly and uniformly distributed. In the present work, the external feedback used by the authors could, in certain parameter regimes, act similarly to periodic driving, potentially leading to a non-uniform phase distribution. Therefore, I would encourage the authors to avoid identifying the CTC phase solely on the basis of full phase circle coverage, and to discuss this subtlety in their manuscript, perhaps referencing the above work for comparison.

Altogether, I find the manuscript to be of high quality, and with the above caveat, I recommend it for publication.

Our response: We sincerely thank the reviewer for the recommendation. We understand the reviewer's concern and appreciate the reference to this important work. It is important to emphasize that our work differs from the discrete time crystal (DTC) regime demonstrated in [Kongkhambut, P. et al., Rep. Prog. Phys. 87, 080502 (2024)]. In our system, the feedback mechanism introduces nonlinear interactions within the system, and there is no periodic external

driving. Therefore, the phase distribution is random and uniform, consistent with the CTC case defined in [Science 377, 670 (2022)] and also with the CTC regime in [Kongkhambut, P. et al., Rep. Prog. Phys. 87, 080502 (2024)]. By contrast, in Ref. [Kongkhambut, P. et al., Rep. Prog. Phys. 87, 080502 (2024)], when the driving (i.e., the pumping) is modulated to be periodic, the phase distribution can become non-uniform or concentrated in particular sectors of the circle, giving rise to a DTC. Considering this, we have added several new sentences in the **Discussion** section in the main text,

on page 5, “*Distinctively, in the discrete time crystal (DTC) scenario, the phase distribution can become non-uniform or concentrated in particular sectors of the circle, as described in Ref. [Kongkhambut, P. et al., Rep. Prog. Phys. 87, 080502 (2024)]. The rich spectrum of dynamical phases...*”.

Reply to Reviewer #4

Comment 4-1

First of all, I would like to thank the authors for engaging in depth with the reports from the reviewers. My assessment of the paper in my first report was positive overall. In reference to the claim of the continuous time crystal (CTC) developing a random oscillation phase, which the authors argue as the distinctive feature of a CTC, I asked the authors to clarify the role played by the feedback activation time, and specifically whether this is short as compared to the CTC period. In their replies, the authors convincingly argue that this is indeed the case. Furthermore, the authors have added to the supplementary information a new Fig S2 showing the time traces of the dynamics, as I requested.

With respect to the other referee reports, I find that the comments from Reviewer 1 have been addressed satisfactorily. The most critical report is arguably that from Reviewer 3. I very much hear their point, namely that CTCs appear as a rebranding of known phenomena in nonlinear dynamical systems. The authors’ reply hinges on the claim that the new and distinctive feature of CTCs is breaking of time translational symmetry, defined as in [Science 377, 6606, 670-673 (2022)], namely in reference to the randomness of the phases of the oscillations. I find this ok, but it is however not textbook knowledge, and should be accordingly emphasized and argued for (for instance, it is currently not mentioned in the discussion, which I invite the authors to do).

Leaving aside the last point, that I leave to Reviewer 3’s judgement, I overall find that the work meets the standards of Nature Communications and could be published there.

Our response: We are grateful to the reviewer for the recommendation and insightful comments. As pointed out by the reviewer, the randomness of the phases of the oscillations, defined by [Science 377, 670 (2022)] as the distinctive feature of CTCs, is not textbook knowledge and is therefore worthy of emphasis. Considering this, we have added several new sentences in the **Discussion** section in the main text,

on page 5, “*...spontaneous symmetry breaking of continuous time translation symmetry that fundamentally distinguishes them. In this work, the criterion for CTCs associated with spontaneous breaking of continuous time-translation symmetry is defined following Ref. [Science 377, 670 (2022)], and is manifested as the random and uniform distribution of oscillation phases.*”.